

SMALL DOLBEAULT EIGENVALUES ALONG HERMITIAN YANG–MILLS FLOW

BERND JOHANNES WUEBBEN

ABSTRACT. We study the small eigenvalues of the Dolbeault Laplacian on trace-free endomorphisms of a semistable holomorphic bundle along Hermitian Yang–Mills flow, normalized as $h^{-1}\partial_\tau h = -2(K_h - \kappa)$ with K_h the mean curvature and κ its average. The bundles considered are simple bundles on a compact curve whose Jordan–Hölder factors are pairwise nonisomorphic, and extensions of one polystable bundle by another on compact Kähler manifolds, again with pairwise nonisomorphic stable factors. From every smooth initial metric, each small eigenvalue is asymptotic to an eigenvalue of a weighted graph Laplacian, whose conductances evolve by an explicit finite-dimensional flow; the relative error tends to zero. The graph records which stable factors are joined by nonzero extension classes. The asymptotics are governed by the weight grading of Haiden, Katzarkov, Kontsevich, and Pandit, the solution of a quadratic program on this graph. Suppose some Lagrange multiplier vector of this program is strictly positive on every active constraint. We then determine the decay exponent and the leading coefficient of every small eigenvalue, and the limits of the spectral subspaces. The eigenvalues of order τ^{-1} have coefficients independent of the initial metric, and $1/2$ is always one of them. At each faster rate the coefficients depend on the initial metric, and their sum is unbounded over metrics on one fixed bundle. A key step shows that the bounded remainder in the asymptotics of the finite-dimensional flow converges. Examples include an eigenvalue decaying like $\tau^{-4/3}$ and, without the positivity condition, an eigenvalue of order $(\tau \log \tau)^{-1}$. We also treat families of harmonic extensions with fixed metrics, including repeated factors and recovery of the graded object, and we show that the constants in a slope inequality relating the lowest eigenvalue to the averaged trace-free mean curvature are sharp in every rank.

1. INTRODUCTION

The summand projections of a polystable direct sum are holomorphic endomorphisms and hence zero modes of its Dolbeault Laplacian. Under a deformation of the holomorphic structure, some of these modes can acquire positive eigenvalues. We ask how the extension data determine the small eigenvalues and their limiting spectral subspaces, and what these limits reveal about the stable summands. Our main concern is the spectrum along Hermitian Yang–Mills flow, where the time parameter is intrinsic rather than chosen. For every simple bundle on a curve whose Jordan–Hölder factors are pairwise nonisomorphic, and for two-step extensions on compact Kähler manifolds, we reduce the small spectrum to a finite-dimensional flow. Under a positivity condition on the associated Lagrange multipliers, we determine the exponent and the leading coefficient of every small eigenvalue, the limits of the spectral subspaces, and which coefficients retain information about the initial metric.

On a compact curve, every stable bundle carries a Hermitian–Einstein metric, whose Chern connection is projectively flat [NS65, Don83], and the Yang–Mills functional was analysed

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from the point of view of Morse theory by Atiyah and Bott [AB83]. Hermitian Yang–Mills flow [Don85, Sim88] carries a semistable bundle on a curve to a Hermitian–Einstein connection on its graded bundle, modulo unitary gauge [Das92, Råd92]. In higher dimensions the limiting reflexive sheaf is the double dual of the graded object of the Harder–Narasimhan–Seshadri filtration [DW04, Jac15, Sib15]. When that graded object is locally free, the convergence is smooth on the whole manifold [SW15].

The exponents are governed by the weight grading of Haiden, Katzarkov, Kontsevich, and Pandit [HKKP23], a solution of a quadratic program on the directed graph of nonzero extension entries. When the Lagrange multipliers of that program are strictly positive, which is the hypothesis of [HKKP23, Theorem 3.6], we show the following. The eigenvalues decaying like τ^{-1} have coefficients that do not depend on the initial metric, and $1/2$ is always one of them. At each faster rate the coefficients depend on the initial metric, and their sum is unbounded over metrics on one fixed bundle. The proof has three parts: a comparison of every small eigenvalue with a finite-dimensional diagonal flow, with relative error tending to zero on the entire future interval; a proof that the bounded remainder in the asymptotics of that flow converges; and a contraction rule for weighted graph Laplacians whose weights decay at different rates. The finite-dimensional models, the exponents, and the logarithmic example come from [HKKP23, HKKP18]. Our contribution is the transfer of precise coefficients to the Dolbeault spectrum of the bundle, the convergence of the remainder, and the resulting dichotomy. Section 1.6 gives the detailed attribution.

The spectral input is a relative comparison for harmonic extension families with fixed background metrics. Scaled harmonic extensions of this kind were used by Sektnan and Tipler in an adiabatic Hermitian Yang–Mills problem [ST24, Section 5]. A finite-dimensional commutator form describes the small spectrum; for pairwise nonisomorphic stable factors it is a weighted graph Laplacian. The error is relative to each eigenvalue, so the estimate remains effective when different modes vanish at different rates.

These asymptotics also give quantitative information about slope stability. The classical subsheaf degree formula [Kob87, Lüb83] yields an inequality relating slopes, the lowest endomorphism eigenvalue, and the trace-free mean curvature. Corrected two-factor families show that its constants are sharp in every rank. Thus the explicit spectral families both describe the approach to a polystable splitting and test the optimality of a general stability estimate.

We use the following conventions throughout. Let (X, ω) be a compact connected Kähler manifold of complex dimension $n \geq 1$, and set

$$dV_\omega = \frac{\omega^n}{n!}, \quad V = \int_X dV_\omega.$$

For a torsion-free coherent sheaf $\mathcal{F}$ of positive rank, define

$$\deg_\omega \mathcal{F} = \int_X c_1(\mathcal{F}) \wedge \frac{\omega^{n-1}}{(n-1)!}, \quad \mu_\omega(\mathcal{F}) = \frac{\deg_\omega \mathcal{F}}{\text{rk } \mathcal{F}}.$$

Here $c_1(\mathcal{F}) = c_1(\det \mathcal{F})$. Following Mumford and Takemoto [Mum63, Tak72], the bundle E is slope-stable if $\mu_\omega(\mathcal{F}) < \mu_\omega(E)$ for every coherent subsheaf with $0 < \text{rk } \mathcal{F} < \text{rk } E$; semistability uses the non-strict inequality; see [Kob87, Chapter V]. It suffices to consider saturated subsheaves, for which $E/\mathcal{F}$ is torsion-free.

For a holomorphic bundle E of rank $r \geq 2$ with a smooth Hermitian metric h , let $\bar{\partial}_{\text{End } E}$ denote the induced Dolbeault operator on $\text{End } E$, and define

$$(1) \quad \lambda_1(h) = \inf_{0 \neq u \in C^\infty(X, \text{End}_0 E)} \frac{\|\bar{\partial}_{\text{End } E} u\|_{L^2}^2}{\|u\|_{L^2}^2}.$$

All L^2 norms use h , ω , and the Hilbert–Schmidt norm on endomorphisms. Thus $\lambda_1(h)$ is the lowest eigenvalue, including zero, of $\bar{\partial}_{\text{End } E}^* \bar{\partial}_{\text{End } E}$ on trace-free endomorphisms; it is not the first positive eigenvalue when the kernel is nonzero. The bundle is simple if $H^0(X, \text{End } E) = \mathbb{C} \text{Id}_E$. By elliptic theory, $\lambda_1(h) > 0$ if and only if E is simple.

1.1. The small spectrum along the flow. Let X be a compact curve, and let E be a simple semistable bundle whose Jordan–Hölder factors $F_1, \dots, F_m$, $m \geq 2$, are pairwise non-isomorphic, of ranks r_i . The *socle* of a semistable bundle is the sum of its stable subbundles of the same slope [HL10, Section 1.5]. The socle filtration $S_1 \subset S_2 \subset \dots \subset E$ takes S_1 to be the socle of E and S_{k+1}/S_k to be the socle of E/S_k . Each F_i occurs in exactly one quotient S_k/S_{k-1} , its *socle layer*. A *two-step extension* is one with two socle layers,

$$0 \longrightarrow S \longrightarrow E \longrightarrow U \longrightarrow 0,$$

with S and U polystable of slope $\mu(E)$.

By Lemma 9.1, E is smoothly identified with $(\bigoplus_i F_i, \bar{\partial}_0 + N)$, where $N = \sum \beta_{ij}$, each $\beta_{ij} \in \Omega^{0,1}(\text{Hom}(F_j, F_i))$ is harmonic, and $\beta_{ij} \neq 0$ only when F_i lies in a lower socle layer than F_j . Let Γ be the directed graph with an edge $i \rightarrow j$ for each nonzero β_{ij} , and put $M_i = V r_i$. We call $\sum_i x_i \text{Id}_{F_i}$, $x_i \in \mathbb{C}$, a *block-scalar* endomorphism. Write $K_h = \sqrt{-1} \Lambda_\omega F_h$ for the mean curvature, and run the normalized metric flow

$$h^{-1} \partial_\tau h = -2(K_h - \kappa \text{Id}), \quad \kappa = 2\pi\mu(E)/V,$$

from any smooth metric. The flow converges to the Hermitian–Einstein connection of the graded bundle [Das92, Råd92]. Exactly $m - 1$ eigenvalues $\lambda_1 \leq \dots \leq \lambda_{m-1}$ of the Dolbeault Laplacian on trace-free endomorphisms therefore tend to zero.

The finite-dimensional comparison object is the *diagonal flow*. It is the flow on block-scalar metrics $g^* h_0$, $g = \text{diag}(e^{y_i} \text{Id}_{F_i})$, in which the mean curvature is replaced by its block-scalar average:

$$(2) \quad \dot{y}_i = -\frac{1}{M_i} \left(\sum_{i \rightarrow j} k_{ij} - \sum_{l \rightarrow i} k_{li} \right), \quad k_{ij} = \|\beta_{ij}\|_{L^2}^2 e^{2(y_i - y_j)}.$$

The edge weight, or *conductance*, k_{ij} is the squared norm of the rescaled extension entry. This is the weighted-graph flow of [HKKP23, (3.10)], and we call y its logarithmic coordinates.

The *weight grading* w of [HKKP23] is the minimizer of $\frac{1}{2} \sum_i M_i w_i^2$ subject to $w_j - w_i \geq \frac{1}{2}$ for every edge $i \rightarrow j$ (Definition 8.1). An edge is *tight* when equality holds, that is, when its constraint is active; the tight edges form the *tight subgraph* $\mathcal{T}$, and its connected components, isolated vertices included, are the *tight components*. The Lagrange multipliers of this program vanish off $\mathcal{T}$. We call $(\Gamma, (r_i))$ *nondegenerate* when some multiplier vector is strictly positive on every tight edge (Definition 8.2). In optimization this is strict complementarity; it is the hypothesis of [HKKP23, Theorem 3.6].

Each edge receives the exponent $a_{ij} = 2(w_j - w_i) \geq 1$. For each value a of these exponents, contract the connected components of the edges with smaller exponent, adding the masses, and join the resulting vertices by the edges of exponent a . Call this the *a-quotient graph*, and let d_a be the number of its positive Laplacian eigenvalues. The small eigenvalues are assigned exponents accordingly: d_a of them receive the exponent a , larger exponents taking smaller indices. Write a_j for the exponent of λ_j .

Theorem 1.1. *Assume $(\Gamma, (r_i))$ is nondegenerate. For every smooth initial metric:*

- (1) $\tau^{a_j} \lambda_j(h(\tau)) \rightarrow c_j(h) > 0$ for every $j < m$.

- (2) The coefficients with $a_j = 1$ are the positive eigenvalues of the Laplacian of $\mathcal{T}$ with masses M_i and conductances u_{ij}^* . Here u^* is the unique multiplier vector of the form $u_{ij}^* = \|\beta_{ij}\|^2 e^{2(z_i - z_j)}$ on the edges of $\mathcal{T}$, with $z \in \mathbb{R}^m$, and zero elsewhere. These coefficients do not depend on the initial metric. The value $\frac{1}{2}$ is always among them; for each tight component $\mathcal{C}$ with at least two vertices it has the eigenvector $\sum_{i \in \mathcal{C}} w_i \text{Id}_{F_i}$.
- (3) For every exponent $a > 1$ with $d_a \geq 1$, the coefficients with $a_j = a$ are the positive eigenvalues of the a -quotient graph with conductances $k_{ij}^\infty(h) = \lim \tau^{a_{ij}} k_{ij}(g_\tau)$. Here g_τ is the diagonal part of $h(\tau)$ in the normal form (Lemma 9.4), and these limits depend on the initial metric. The sum of the coefficients is unbounded as the initial metric varies on E .
- (4) In unitary gauges in which the flow converges, the spectral projection onto the eigenvalues with a given exponent and a given coefficient converges smoothly. The limit is the corresponding eigenspace of the Laplacian of the a -quotient graph ($\mathcal{T}$ when $a = 1$), embedded as block-scalar endomorphisms constant on the contracted vertices.

The same holds for two-step extensions on compact Kähler manifolds of any dimension, when the factors are vector bundles.

Theorem 1.1 combines Theorems 10.1 and 12.1. Its basis is a comparison that needs no nondegeneracy (Theorem 9.5): for every sufficiently large T there are a solution $g^{[T]}$ of (2) on $[T, \infty)$ and $\delta_T \rightarrow 0$ with

$$e^{-\delta_T} \nu_j(g^{[T]}(\tau)) \leq \lambda_j(h(\tau)) \leq e^{\delta_T} \nu_j(g^{[T]}(\tau)), \quad \tau \geq T,$$

where the ν_j are the positive eigenvalues of the weighted graph Laplacian with masses M_i and conductances k_{ij} . Without nondegeneracy, iterated logarithms can appear in the diagonal flow.

Three examples show the range of behaviour.

- For a nonsplit extension of two stable bundles, $\tau \lambda_1(h(\tau)) \rightarrow \frac{1}{2}$. The limit is independent of the ranks, the extension class, the initial metric, and the dimension (Theorems 7.1 and 12.1).
- Take four stable factors of ranks $(2, 1, 1, 2)$ with nonzero entries exactly on the edges $1 \rightarrow 2, 3 \rightarrow 2, 3 \rightarrow 4$ (a zigzag). Two eigenvalues are asymptotic to $1/(2\tau)$, and $\lambda_1 \sim c(h(0))\tau^{-4/3}$. With $V = 1$, unit-norm entries, and the initial metrics $h_\rho = g_\rho^* h_0$ with $g_\rho = \text{diag}(\rho^{1/2}, \rho^{-1/2}, \rho^{1/2}, \rho^{-1/2})$, one has $c(h_\rho) = \frac{2}{27}\rho^{-2/3}(1 + O(\rho))$ (Proposition 11.3).
- For four line bundles with the same zigzag, the multiplier on the edge $3 \rightarrow 2$ is forced to vanish, so $(\Gamma, (r_i))$ is degenerate. Here $\lambda_1 \sim (2\tau \log \tau)^{-1}$, an eigenvalue separated from the others by a logarithmic factor, and $\lambda_2 \sim \lambda_3 \sim (2\tau)^{-1}$ (Theorem 11.4).

The exponent $4/3$ and the logarithmic model are features of the weighted graph flows of [HKKP23, Section 3.2 and Example 5.13]. The theorems here determine the corresponding Dolbeault coefficients along the actual flow.

The finite-dimensional input is Theorem 8.3. For nondegenerate $(\Gamma, (r_i))$, every solution of (2) satisfies

$$y(\tau) = w \log \tau + \zeta^* + \sum_{\mathcal{C}} Z_{\mathcal{C}} \mathbf{1}_{\mathcal{C}} + O(\tau^{-\delta}).$$

Here ζ^* is the minimizer of an explicit strictly convex function on the tight subgraph, determined by Γ , the ranks, V and the norms $\|\beta_{ij}\|$; $\mathbf{1}_{\mathcal{C}}$ is the indicator vector of the tight component $\mathcal{C}$. The constants $Z_{\mathcal{C}}$ depend on the solution. This refines the bounded remainder

of [HKKP23, Theorem 3.6]. It also yields convergence of the bounded term in the metric asymptotics along the actual flow (Corollary 10.2). The constants Z_C are exactly what makes the faster coefficients depend on the initial metric.

1.2. Harmonic extensions and their small spectrum. First suppose that X is a compact curve. Fix pairwise nonisomorphic stable bundles $F_1, \dots, F_m$, $m \geq 2$, of equal slope and ranks r_i , and equip $E_0 = \bigoplus_i F_i$ with a product Hermitian–Einstein metric. Choose harmonic forms $\beta_{ij} \in \Omega^{0,1}(\text{Hom}(F_j, F_i))$, $i < j$, at least one of which is nonzero. For strictly decreasing integers $\ell_1 > \dots > \ell_m = 0$, set

$$p_{ij} = \ell_i - \ell_j, \quad \bar{\partial}_t = \bar{\partial}_0 + \sum_{i < j} t^{p_{ij}} \beta_{ij}, \quad t > 0.$$

Each entry occupies its indicated upper-triangular block. Integrability is automatic on a curve; denote the resulting holomorphic bundle by E_t . The diagonal transformation $g_t = \text{diag}(t^{\ell_i} \text{Id}_{F_i})$ identifies all positive parameters with the same holomorphic bundle. We retain the fixed product metric when computing the spectrum. Equivalently, g_t transports it to a family of metrics on that fixed holomorphic bundle.

The split trace-free holomorphic endomorphisms form the space

$$\mathcal{K} = \left\{ \text{diag}(x_i \text{Id}_{F_i}) : \sum_i r_i x_i = 0 \right\}.$$

Associate an undirected graph to the nonzero entries β_{ij} , with vertex masses $V r_i$ and edge conductances $t^{2p_{ij}} \|\beta_{ij}\|_{L^2}^2$. Its weighted Laplacian L_t on $\mathcal{K}$ is defined by

$$\langle x, L_t x \rangle = \sum_{i < j} t^{2p_{ij}} \|\beta_{ij}\|_{L^2}^2 |x_i - x_j|^2, \quad \|x\|^2 = V \sum_i r_i |x_i|^2.$$

Let $\nu_1(t) \leq \dots \leq \nu_{m-1}(t)$ be its eigenvalues and $\lambda_1(t) \leq \lambda_2(t) \leq \dots$ those of the Dolbeault Laplacian on trace-free endomorphisms of E_t , counting complex multiplicity and including zero. Put $p_* = \min_{\beta_{ij} \neq 0} p_{ij}$. The following statement is proved in Section 3.

Theorem 1.2 (Spectral comparison). *Exactly $m - 1$ eigenvalues tend to zero as $t \rightarrow 0$, and the remaining spectrum stays uniformly bounded away from zero. For every positive $\nu_j(t)$,*

$$\frac{\lambda_j(t)}{\nu_j(t)} = 1 + O(t^{2p_*}),$$

with an error uniform in j for the fixed family. If the graph has c connected components, exactly $c - 1$ of the small eigenvalues are identically zero. Let Π_t be the orthogonal projection onto the full small spectral subspace and P the orthogonal projection onto $\mathcal{K}$. For every $k \geq 0$,

$$\|\Pi_t - P\|_{L^2 \rightarrow C^k} = O(t^{2p_*}).$$

Theorem 4.1 determines every leading decay order by a finite construction. At an edge exponent p , contract the components joined by edges with $p_{ij} < p$ and add their vertex masses. Join the resulting vertices using entries with $p_{ij} = p$, assigning conductances $\|\beta_{ij}\|_{L^2}^2$ and adding contributions between the same pair of vertices. The positive eigenvalues ρ of this quotient graph give precisely the coefficients of the modes $\lambda(t) = t^{2p}(\rho + o(1))$. Their spectral subspaces converge smoothly to the corresponding quotient eigenspaces, embedded as endomorphisms constant on each contracted component. For a repeated coefficient, the conclusion concerns the whole eigenspace. For the matrix family L_t alone, the contraction rule

follows from the reduction process of analytic perturbation theory [Kat95, Chapter II, §2.3]; Theorem 4.1 transfers it, together with the limiting subspaces, to the Dolbeault spectrum.

The limiting full subspace has a geometric interpretation. After adjoining the scalar identity to $\mathcal{K}$, one obtains the algebra $\bigoplus_i \mathbb{C} \text{Id}_{F_i}$. Its minimal nonzero idempotents are the projections onto the stable summands. The limiting spectral subspace, viewed as a space of endomorphisms with pointwise multiplication, therefore identifies the unordered summands of this split limit.

For a concrete example, take three pairwise nonisomorphic degree-zero line bundles on a curve of genus at least two and volume one. Choose harmonic forms of unit L^2 norm on the consecutive blocks 12, 23, scale them by t, t^2 , and set the 13 entry to zero. The two small eigenvalues satisfy

$$\lambda_1(t) = \frac{3}{2}t^4(1 + O(t^2)), \quad \lambda_2(t) = 2t^2(1 + O(t^2)).$$

The limiting eigensection for the smaller eigenvalue distinguishes $F_1 \oplus F_2$ from F_3 ; that for the larger eigenvalue distinguishes F_1 from F_2 . An absolute error of order t^4 would obscure the first leading coefficient, whereas the relative comparison determines both. Section 5 derives these formulas and gives a five-factor example with a repeated coefficient.

1.3. Realization for semistable bundles on curves. The harmonic families can be constructed from every simple strictly semistable bundle E on a compact curve. Choose a Jordan–Hölder filtration with stable quotients $F_1, \dots, F_m$, allowing repetitions. Its graded bundle $G = \text{gr}(E) = \bigoplus_i F_i$ is determined up to isomorphism by E [Ses67]; see also [HL10, Section 1.5]. After choosing Hermitian–Einstein metrics on the quotients, compatible with fixed identifications between isomorphic copies, Lemma 6.1 gives a filtration-preserving smooth identification in which all upper-triangular extension entries are harmonic. The construction corrects successive upper diagonals and retains the changes they induce in higher entries. It requires no smallness assumption on the original extension.

Write $G = \bigoplus_\alpha H_\alpha \otimes \mathbb{C}^{m_\alpha}$ with pairwise nonisomorphic stable H_α . Since stable bundles are simple and a nonzero homomorphism between stable bundles of equal slope is an isomorphism [Kob87, Chapter V, Proposition 7.11 and Corollaries 7.12, 7.14], its holomorphic endomorphism algebra is $\bigoplus_\alpha M_{m_\alpha}(\mathbb{C})$. Theorem 13.1 extends the relative comparison to the quadratic form below, where N_t is the harmonic extension form after diagonal scaling:

$$\langle x, L_t x \rangle = \|[N_t, x]\|_{L^2}^2, \quad x \in H^0(X, \text{End}_0 G).$$

Theorem 14.1 constructs approximate Hermitian–Einstein metrics h_t (cf. [Kob87, Chapter VI, Theorem 10.13] and [Jac14]) and explicit gauges (79) in which this comparison applies. Exactly $d = \sum_\alpha m_\alpha^2 - 1$ positive eigenvalues tend to zero, the remaining spectrum has a uniform gap, and the full small spectral projection converges smoothly to the projection onto $H^0(X, \text{End}_0 G)$. Adjoining the scalar identity recovers the endomorphism algebra of the graded bundle. Its central idempotents identify the isotypical components, its matrix sizes give the multiplicities, and the images of its minimal idempotents give the stable factors up to isomorphism. Individual repeated copies have no canonical projections. For example, Section 13.2 constructs a simple bundle with graded object $A \oplus B \oplus A \oplus C$ and five small modes, with leading terms $t^4, t^2, t^2, 2t^2, 2t^2$. These recovery statements concern the constructed metric families.

1.4. Slope inequalities and their sharpness. We now return to an arbitrary smooth Hermitian metric h on a bundle E over a compact connected Kähler manifold. The implication from irreducible Hermitian–Einstein metrics to slope stability rests on the subbundle

curvature formula and its extension to coherent subsheaves [Kob87, Lüb83]. The converse is the Donaldson–Uhlenbeck–Yau theorem [Don85, UY86]. For a general metric, the same curvature formula gives a quantitative estimate with an error term.

Write F_h for its Chern curvature, with the convention $c_1(E, h) = \frac{\sqrt{-1}}{2\pi} \operatorname{tr} F_h$, and put

$$(3) \quad K_h = \sqrt{-1} \Lambda_\omega F_h, \quad K_h^0 = K_h - \frac{\operatorname{tr} K_h}{r} \operatorname{Id}_E.$$

Both endomorphisms are self-adjoint with respect to h . The quantity used to measure the curvature error is

$$(4) \quad \varepsilon_1(h) = \frac{1}{V} \int_X \|K_h^0\|_{\operatorname{op}} dV_\omega.$$

Theorem 1.3. *For every saturated coherent subsheaf $\mathcal{S} \subset E$ of rank $0 < s < r$, one has*

$$(5) \quad \mu_\omega(E) - \mu_\omega(\mathcal{S}) \geq \frac{V(r-s)}{2\pi r} \left(\lambda_1(h) - \frac{r}{\max\{s, r-s\}} \varepsilon_1(h) \right).$$

In particular, if

$$(6) \quad \lambda_1(h) > c_r \varepsilon_1(h), \quad c_r = \frac{r}{\lceil r/2 \rceil},$$

then E is slope-stable.

The proof combines the classical subsheaf degree formula with the Rayleigh principle; we have not found the inequality stated in this form. Its test section is the orthogonal projection onto the subsheaf with its scalar part removed; its pointwise squared norm is $s(r-s)/r$. A trace-free matrix estimate gives the displayed rank dependence. Since $c_r \leq 2$ and $\varepsilon_1(h) \leq \|K_h^0\|_{L^\infty, \operatorname{op}}$, the simpler condition $2\|K_h^0\|_{L^\infty, \operatorname{op}} < \lambda_1(h)$ also implies stability. No simplicity assumption is needed: the strict inequality already implies $\lambda_1(h) > 0$.

A strictly semistable bundle has a proper saturated subsheaf of equal slope. Theorem 1.3 therefore forces $\lambda_1(h_j) \rightarrow 0$ whenever $\varepsilon_1(h_j) \rightarrow 0$, in particular along approximate Hermitian–Einstein metrics. This conclusion concerns arbitrary such sequences; it does not require the harmonic form of the families above.

The relation to the explicit spectral asymptotics is clearest for two factors. Let F, G be nonisomorphic stable bundles of equal slope on a compact Kähler manifold, of ranks a, b , with fixed Hermitian–Einstein metrics and a nonzero harmonic extension form $\beta \in \Omega^{0,1}(\operatorname{Hom}(G, F))$. Scaling the extension by t gives exactly one small eigenvalue. Theorem 2.1 gives

$$\lambda_1(t) = \frac{(a+b)\|\beta\|_{L^2}^2}{Vab} t^2 - dt^4 + O(t^6), \quad d \geq 0,$$

with an explicit formula for d in terms of the inverse split Laplacian on the complement of its kernel. In the gauge g_t of (16), the normalized eigensection converges smoothly to the normalized trace-free part of a summand projection. This two-factor result holds in every complex dimension.

A second-order correction of the metric, by the Green operator applied to the trace-free part of the curvature, makes both estimates in the proof of Theorem 1.3 asymptotically sharp. On a curve of genus at least two, for every $r \geq 2$ and $1 \leq s < r$, this gives nonsplit extensions of a stable bundle of rank $r-s$ by one of rank s , and metrics with

$$\frac{\lambda_1}{\varepsilon_1} \longrightarrow \frac{r}{\max\{s, r-s\}}$$

(Corollary 15.9). These bundles are strictly semistable, so no coefficient in (5) can be decreased, and c_r is optimal in every rank. The same families show that c_r is optimal for the criterion with $\|K_h^0\|_{L^\infty, \text{op}}$ in place of ε_1 .

1.5. Organization. Sections 2–5 treat harmonic families with fixed metrics. They contain the two-factor expansion, the relative comparison with the graph Laplacian, and the contraction rule for decay orders. Section 6 gives the harmonic normal form on curves. Sections 7–12 contain the flow results.

- Section 7 treats two factors directly.
- Section 8 proves the sharp asymptotics of the diagonal flow.
- Section 9 compares the bundle flow with diagonal flows begun at arbitrarily late times.
- Section 10 proves Theorem 1.1, Section 11 gives the examples, and Section 12 treats higher dimensions.

The comparison at late times is essential. A comparison begun at a single fixed time gives only two-sided bounds of order τ^{-a} , not the coefficients. Sections 13 and 14 return to fixed metrics, with repeated factors and a recovery theorem. The slope inequality is proved at the end.

1.6. Relation to earlier work and scope. The scaled-extension construction is classical in this setting. Sektnan–Tipler [ST24, Section 5] use it in an adiabatic Hermitian Yang–Mills problem; their Proposition 41 computes the action on summand identities, and Lemma 52 uses a weighted quadratic form in their differences. Their linearized operator is a connection Laplacian with a varying base metric. Here we work with the nonnegative Dolbeault Laplacian and fixed metrics, derive a relative comparison for each small eigenvalue, and prove convergence of the full small spectral subspace. For pairwise nonisomorphic factors, graph contraction also determines every leading coefficient and its limiting subspace. The two-factor expansion uses standard analytic perturbation theory [Kat95, Chapter VII].

For the flow theorems, existence and convergence to the graded bundle are classical inputs [Das92, Råd92]; in higher dimensions we use Sibley and Wentworth’s identification of the singular set [SW15, Theorem 1.1]. Haiden, Katzarkov, Kontsevich, and Pandit [HKKP18] describe metric growth along the flow on curves by weight filtrations and iterated logarithms, with a bounded remainder. Their weighted-graph analysis [HKKP23, Section 3] supplies the weight grading, the multiplier condition, the convexity argument, the regime giving the exponent $4/3$, and the logarithmic model of Example 5.13. Our proofs use their Green operator correction and order comparison [HKKP18, Sections 3.5 and 4.2].

Lam [Lam26, Theorems 1.1 and 8.2, Proposition 8.6] extends the metric asymptotics to compact Kähler manifolds with locally free graded object, including repeated factors, again with a bounded remainder. A fixed multiplicative bound between metrics leaves a fixed uncertainty in each leading coefficient. Theorem 9.5 removes that uncertainty by comparing with diagonal metrics begun at arbitrarily late times. The convergence of the remainder in the diagonal flow, Theorem 8.3, then determines the coefficients. That convergence also sharpens the metric asymptotics in the nondegenerate case.

After a linear change of variables, the diagonal flow is gradient flow for an exponential loss with linearly separable data, and the weight grading is the hard-margin direction (Remark 8.4). For gradient descent with unit weights, Soudry, Hoffer, Nacson, Gunasekar, and Srebro [SHNGS18] prove convergence of the remainder when the support vectors span the data. Here that means there is a single tight component. Nacson et al. [NLGSSS19, Theorem 10] bound the component of the remainder outside the span of the support vectors

without that assumption. In our setting there can be several tight components; the argument for convergence in that case is short. The dependence on the initial metric enters exactly through those components.

The harmonic gauge condition itself also appears in [ST24, Section 5.1]; our triangular construction on a curve supplies the normal form needed to apply the spectral comparison to the bundle E .

We state the scope of the fixed-metric results once. They concern the constructed families: harmonic extension data, a chosen filtration, quotient metrics, and scaling exponents, with rates referring to the chosen parameter t . The flow results use the intrinsic time τ . The identification of stable factors uses the limiting space of endomorphisms with its multiplication, not the eigenvalue list alone, and it does not select a canonical Jordan–Hölder filtration. For arbitrary approximate Hermitian–Einstein sequences, the slope inequality gives eigenvalue collapse, but convergence of spectral subspaces is not asserted.

2. AN EXTENSION OF TWO STABLE BUNDLES

Let (X, ω) again be a compact connected Kähler manifold. Let F, G be nonisomorphic stable bundles of equal slope, with ranks a, b , and put $r = a + b$. Fix Hermitian–Einstein metrics h_F, h_G , whose common Einstein constant is $\kappa = 2\pi\mu_\omega(F)/V$, and a nonzero harmonic form

$$\beta \in \Omega^{0,1}(\text{Hom}(G, F)), \quad \bar{\partial}\beta = 0, \quad \bar{\partial}^*\beta = 0.$$

All inner products in this section use $h_0 = h_F \oplus h_G$ and the fixed Kähler form. We use the convention $\langle u, v \rangle = \int_X \text{tr}(u^*v) dV_\omega$.

On the fixed smooth bundle $F \oplus G$ define

$$(7) \quad N = \begin{pmatrix} 0 & \beta \\ 0 & 0 \end{pmatrix}, \quad \bar{\partial}_t = \bar{\partial}_0 + tN, \quad t \in \mathbb{R}.$$

This is integrable in every complex dimension because $\bar{\partial}_0 N = 0$ and $N \wedge N = 0$. Denote the resulting holomorphic bundle by E_t . On trace-free endomorphisms put

$$D_t = D_0 + tB, \quad D_0 = \bar{\partial}_{0, \text{End}}, \quad Bu = [N, u], \quad \Delta_t = D_t^* D_t.$$

These nonnegative self-adjoint elliptic operators have the fixed domain H^2 in L^2 . Eigenvalues count complex multiplicity and include zero. Let P_F be the orthogonal projection onto F and set

$$(8) \quad q = P_F - \frac{a}{r} \text{Id} = \text{diag} \left(\frac{b}{r} \text{Id}_F, -\frac{a}{r} \text{Id}_G \right), \quad V_q = \|q\|_{L^2}^2 = \frac{Vab}{r}.$$

Write $\beta = \sum_\nu \beta_\nu \bar{\theta}^\nu$ in a local unitary coframe, and define the global self-adjoint trace-free endomorphism

$$(9) \quad C = \text{diag} \left(\sum_\nu \beta_\nu \beta_\nu^*, -\sum_\nu \beta_\nu^* \beta_\nu \right).$$

Stability gives $\ker \Delta_0 = \mathbb{C}q$, as verified below. Let P be the L^2 projection onto this kernel, $Q = \text{Id} - P$, and $R = (\Delta_0|_{QH^2})^{-1} : QL^2 \rightarrow QH^2$. Define

$$(10) \quad c = \frac{r\|\beta\|_{L^2}^2}{Vab}, \quad z = C - cq = QC, \quad d = \frac{\langle z, Rz \rangle}{V_q}.$$

Theorem 2.1. *For the family (7), E_t is simple and strictly semistable for every $t \neq 0$. Exactly one eigenvalue of Δ_t tends to zero, and*

$$(11) \quad \lambda_1(t) = ct^2 - dt^4 + O(t^6), \quad c > 0, \quad d \geq 0.$$

There are $\delta, t_0 > 0$ such that $\lambda_2(t) \geq \delta$ for $|t| < t_0$. The normalized first eigensection, with phase chosen so that $\langle q, u_t \rangle > 0$, satisfies for every $k \geq 0$

$$(12) \quad u_t = \frac{q}{\sqrt{V_q}} - \frac{t^2}{\sqrt{V_q}} Rz + O_{C^k}(t^3).$$

The mean curvature and its averaged trace-free norm are exactly

$$(13) \quad K_t = \kappa \text{Id} + t^2 C, \quad \varepsilon_1(t) = \frac{t^2}{V} \int_X \|C\|_{\text{op}} dV_\omega.$$

Moreover, $\lambda_1(t) \leq ct^2$ for every real t .

Proof. Nonisomorphism and stability at equal slope imply

$$H^0(\text{Hom}(F, G)) = H^0(\text{Hom}(G, F)) = 0, \quad H^0(\text{End } F) = \mathbb{C} \text{Id}_F, \quad H^0(\text{End } G) = \mathbb{C} \text{Id}_G.$$

Thus the trace-free holomorphic endomorphisms of the split bundle are precisely $\mathbb{C}q$. This proves the assertion about $\ker \Delta_0$ and the existence of R .

For $t \neq 0$, an endomorphism of E_t preserves F , since its composite $F \rightarrow E_t \rightarrow G$ vanishes. It induces scalars α, η on F, G , and its upper-right block v satisfies

$$\bar{\partial}v + t(\eta - \alpha)\beta = 0.$$

Pairing with β gives $\eta = \alpha$. The block v is then holomorphic and hence zero, proving simplicity. For any subsheaf of E_t , its intersection with F and image in G have slopes at most the common slope whenever their ranks are nonzero. Degree and rank additivity prove semistability, and the equal-slope subbundle F rules out stability.

Expand on the common domain:

$$(14) \quad \Delta_t = \Delta_0 + tA_1 + t^2A_2, \quad A_1 = D_0^*B + B^*D_0, \quad A_2 = B^*B.$$

Since $D_0q = 0$, $Bq = -N$, and $D_0^*N = 0$ by harmonicity, one has $A_1q = 0$. The adjoint formula $B^*v = \sum_\nu [N_\nu^*, v_\nu]$ gives

$$(15) \quad A_2q = C, \quad \Delta_tq = t^2C, \quad \langle q, C \rangle = \|\beta\|_{L^2}^2.$$

In particular $PC = cq$, and the Rayleigh quotient of q gives $\lambda_1(t) \leq ct^2$.

The zero eigenvalue of Δ_0 is isolated and simple. The analytic family (14) has compact resolvent and constant domain, so its spectral projection near zero is analytic and of rank one for small $|t|$ [Kat95, Chapter VII]. The remaining spectrum stays bounded away from zero. For clarity, on a contour separating zero from the positive spectrum, $(\Delta_0 - \zeta)^{-1}$ maps L^2 to H^2 , and $(tA_1 + t^2A_2)(\Delta_0 - \zeta)^{-1}$ has norm $O(|t|)$ on L^2 . A Neumann series constructs the perturbed resolvent and its contour integral. The same argument between H^s and H^{s+2} gives Taylor expansions of the eigensection in every C^k norm.

Choose the analytic eigensection $v(t)$ with $\langle q, v(t) \rangle = V_q$ and write

$$v(t) = q + tv_1 + t^2v_2 + t^3v_3 + \cdots, \quad v_j \perp q.$$

The order- t equation gives $\lambda'(0) = 0$ and $v_1 = 0$. At order t^2 , projection onto q and $q^\perp$ gives respectively the coefficient c and

$$\Delta_0v_2 + C = cq, \quad v_2 = -Rz.$$

The order- t^3 coefficient vanishes because $A_1 q = 0$. If $\lambda_{[4]}$ denotes the coefficient of t^4 , then

$$V_q \lambda_{[4]} = \langle q, A_1 v_3 + A_2 v_2 \rangle = -\langle z, Rz \rangle.$$

This coefficient is $-d$, with $d \geq 0$ because R is positive. Conjugation by $J = \text{diag}(-\text{Id}_F, \text{Id}_G)$ is unitary and intertwines D_t with D_{-t} . Uniqueness of the isolated branch makes its eigenvalue even in t , proving the $O(t^6)$ remainder in (11). Finally, $\|v(t)\|^2 = V_q + O(t^4)$, so normalization gives (12).

The Chern connection is $\nabla_t = \nabla_0 + t(N - N^*)$, where the adjoint also changes the form type. The Kähler identity $\bar{\partial}^* \beta = -\sqrt{-1} \Lambda_\omega \partial \beta = 0$ and its adjoint kill the linear term in the contracted curvature. In a unitary coframe with $\omega = \sqrt{-1} \sum \theta^\nu \wedge \bar{\theta}^\nu$, the quadratic term is

$$\sqrt{-1} \Lambda_\omega (N - N^*) \wedge (N - N^*) = C.$$

Its trace is zero, proving (13). $\square$

To view this degeneration on the fixed holomorphic bundle E_1 , set $g_t = \text{diag}(t \text{Id}_F, \text{Id}_G)$ for $t > 0$. Then

$$(16) \quad \bar{\partial}_t = g_t \bar{\partial}_1 g_t^{-1}, \quad h_t = g_t^* h_0 = t^2 h_F \oplus h_G.$$

The map $g_t : (E_1, h_t) \rightarrow (E_t, h_0)$ is a holomorphic isometry and identifies the endomorphism Laplacians unitarily. Thus h_t is an approximate Hermitian–Einstein family on E_1 . The convergence in (12) is in the gauge g_t of (16). Its limit u_0 has two distinct fiberwise eigenvalues, and

$$P_F = \sqrt{V_q} u_0 + \frac{a}{r} \text{Id}, \quad P_G = \text{Id} - P_F.$$

3. SEVERAL STABLE SUMMANDS ON A CURVE

Let (X, ω) now be a compact connected Riemann surface. Fix pairwise nonisomorphic stable bundles $F_1, \dots, F_m$, $m \geq 2$, of equal slope, with ranks r_i and Hermitian–Einstein metrics h_i . On $E_0 = \bigoplus_i F_i$ use the metric $h_0 = \bigoplus_i h_i$. Choose strictly decreasing integers $\ell_1 > \dots > \ell_m = 0$, the *scaling exponents*, and harmonic forms

$$\beta_{ij} \in \Omega^{0,1}(\text{Hom}(F_j, F_i)), \quad i < j.$$

Some forms may vanish, but at least one is nonzero. Set

$$(17) \quad p_{ij} = \ell_i - \ell_j, \quad N_t = \sum_{i < j} t^{p_{ij}} \beta_{ij}, \quad \bar{\partial}_t = \bar{\partial}_0 + N_t, \quad t > 0,$$

where each summand occupies its indicated upper-triangular block. Integrability is automatic on a curve. The diagonal transformation $g_t = \text{diag}(t^{\ell_i} \text{Id}_{F_i})$ satisfies $\bar{\partial}_t = g_t \bar{\partial}_1 g_t^{-1}$. Thus all E_t , $t > 0$, are holomorphically isomorphic, and g_t is an isometry from $(E_1, g_t^* h_0)$ to (E_t, h_0) .

On the fixed Hilbert space $H = L^2(X, \text{End}_0 E_0)$ define

$$D_t = D_0 + B_t, \quad D_0 = \bar{\partial}_{0, \text{End}}, \quad B_t u = [N_t, u], \quad \Delta_t = D_t^* D_t.$$

The form domain is H^1 and the operator domain is H^2 . Stability and nonisomorphism give

$$(18) \quad \mathcal{K} = \ker D_0 = \left\{ \text{diag}(x_i \text{Id}_{F_i}) : \sum_i r_i x_i = 0 \right\}, \quad \dim_{\mathbb{C}} \mathcal{K} = m - 1.$$

Let $P : H \rightarrow \mathcal{K}$ be the orthogonal projection, $Q = \text{Id} - P$, and let $\gamma > 0$ be the lowest eigenvalue of Δ_0 on QH . Put

$$\eta_t = \|B_t\|_{L^2 \rightarrow L^2(\Omega^{0,1})}, \quad p_* = \min_{\beta_{ij} \neq 0} p_{ij}.$$

The coefficients of B_t are $O(t^{p*})$ in every differentiability norm; in particular $\eta_t = O(t^{p*})$.

3.1. The weighted graph and the relative estimate. Associate to (17) the undirected graph with vertices $1, \dots, m$ and edges ij for which $\beta_{ij} \neq 0$. Define its vertex masses and conductances by

$$(19) \quad M_i = Vr_i, \quad c_{ij} = \|\beta_{ij}\|_{L^2}^2, \quad k_{ij}(t) = c_{ij}t^{2p_{ij}}.$$

Absent conductances are zero. The weighted graph Laplacian L_t on $\mathcal{K}$ is defined by

$$(20) \quad A_t(x) := \langle x, L_t x \rangle = \sum_{i < j} k_{ij}(t) |x_i - x_j|^2, \quad \|x\|^2 = \sum_i M_i |x_i|^2.$$

Equivalently, $(L_t x)_i = M_i^{-1} \sum_{j \neq i} k_{ij}(t)(x_i - x_j)$, restricted to $\sum_i M_i x_i = 0$. Write $\nu_1(t) \leq \dots \leq \nu_{m-1}(t)$ for its eigenvalues and $\lambda_1(t) \leq \lambda_2(t) \leq \dots$ for those of Δ_t on H . Both lists count multiplicity and include zero.

Theorem 3.1. *If $\eta_t \leq \sqrt{\gamma}/4$ and $\gamma_t = (\sqrt{\gamma} - \eta_t)^2$, then*

$$(21) \quad \left(1 - \frac{2\eta_t^2}{\gamma_t}\right) \nu_j(t) \leq \lambda_j(t) \leq \nu_j(t), \quad 1 \leq j \leq m-1, \quad \lambda_m(t) \geq \frac{\gamma}{4}.$$

There are exactly $m-1$ eigenvalues below $\gamma/4$, including zeros. For every positive graph eigenvalue,

$$(22) \quad \frac{\lambda_j(t)}{\nu_j(t)} = 1 + O(t^{2p*}).$$

The error is uniform in j for this fixed family, regardless of the ratios between its positive graph eigenvalues. If the graph has c connected components, then Δ_t has exactly $c-1$ zero eigenvalues on trace-free endomorphisms for every $t > 0$. In particular, E_t is simple if and only if the graph is connected.

Proof. For $x \in \mathcal{K}$, the ij block of $B_t x$ is $t^{p_{ij}}(x_j - x_i)\beta_{ij}$. The different blocks are orthogonal, so

$$\|B_t x\|^2 = A_t(x).$$

Harmonicity gives the cancellation

$$(23) \quad D_0^* B_t x = 0 \quad (x \in \mathcal{K}).$$

For $y \in QH^1$, the split spectral gap gives

$$(24) \quad \|D_t y\| \geq \|D_0 y\| - \|B_t y\| \geq (\sqrt{\gamma} - \eta_t) \|y\| = \sqrt{\gamma_t} \|y\|.$$

Decompose $u = x + y$ with $x \in \mathcal{K}$, $y \in QH^1$. By (23),

$$\|D_t u\|^2 = A_t(x) + 2 \operatorname{Re} \langle B_t x, B_t y \rangle + \|D_t y\|^2.$$

The cross term has absolute value at most $2\eta_t A_t(x)^{1/2} \|y\|$. Young's inequality therefore gives

$$(25) \quad \|D_t u\|^2 \geq \left(1 - \frac{2\eta_t^2}{\gamma_t}\right) A_t(x) + \frac{\gamma_t}{2} \|y\|^2.$$

The min-max principle on subspaces of $\mathcal{K}$ gives the upper bound in (21). For the lower bound, compare with the orthogonal sum of $(1 - 2\eta_t^2/\gamma_t)L_t$ on $\mathcal{K}$ and $(\gamma_t/2)\operatorname{Id}$ on QH . Its variational values below $\gamma_t/2$ are the graph eigenvalues with the indicated prefactor. Since

$$\nu_{m-1}(t) \leq \eta_t^2 \leq \gamma/16, \quad \gamma_t/2 \geq 9\gamma/32,$$

min–max proves (21), including $\lambda_m(t) \geq \gamma_t/2 > \gamma/4$. This also proves (22); its constants may depend on the fixed bundles and forms.

The kernel of the graph form consists of vectors constant on each connected component. Its intersection with $\mathcal{K}$ has dimension $c - 1$. Formula (25) shows that these are exactly the zero modes of Δ_t for small t . The positive-parameter complex gauge equivalence gives the same kernel dimension for all $t > 0$. Adding the scalar identity proves the assertion about simplicity. $\square$

Remark 3.2. The cancellation (23) is needed. Without it, complementary directions can change the leading term of the form restricted to $\mathcal{K}$. Already in finite dimensions, $D_0 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $B_t = \begin{pmatrix} t & 0 \\ 0 & 0 \end{pmatrix}$ have restricted energy t^2 on $\ker D_0$, while $(D_0 + B_t)^*(D_0 + B_t)$ has $(-1, t)$ in its kernel. So the relative comparison does not follow from smallness of B_t alone.

3.2. The complementary equation and smooth convergence. On QH let $T_t = Q\Delta_t Q$, with domain QH^2 , be the operator of the restricted closed quadratic form. It satisfies $T_t \geq \gamma_t$, where γ_t is as in Theorem 3.1. By (23), the off-diagonal block is

$$(26) \quad R_t = Q\Delta_t P = QB_t^* B_t P, \quad \|R_t x\| \leq \eta_t A_t(x)^{1/2}.$$

For $0 \leq \lambda < \gamma_t/2$, solving the Q component of the eigenvalue equation gives

$$(27) \quad y = -(T_t - \lambda)^{-1} R_t x, \quad S_t(\lambda)x = \lambda x, \quad S_t(\lambda) = L_t - R_t^*(T_t - \lambda)^{-1} R_t.$$

The correction in this Schur complement satisfies the relative form bound

$$(28) \quad 0 \leq L_t - S_t(\lambda) \leq \frac{2\eta_t^2}{\gamma_t} L_t.$$

Cauchy–Schwarz for the positive form on the left yields, for $x, z \in \mathcal{K}$,

$$(29) \quad |\langle z, (L_t - S_t(\lambda))x \rangle| \leq \frac{2\eta_t^2}{\gamma_t} A_t(z)^{1/2} A_t(x)^{1/2}.$$

Proposition 3.3. *For every $k \geq 0$, a unit eigensection u belonging to one of the $m - 1$ small eigenvalues λ satisfies*

$$(30) \quad \|Qu\|_{C^k} \leq C_k t^{p^*} \sqrt{\lambda}.$$

If Π_t denotes the orthogonal projection onto the full small spectral subspace, then

$$(31) \quad \|\Pi_t - P\|_{L^2 \rightarrow C^k} = O(t^{2p^*}).$$

Proof. Write $u = x + y$ as above and set $\epsilon_t = 2\eta_t^2/\gamma_t$. Formula (25) implies $A_t(x) \leq \lambda/(1 - \epsilon_t)$. Equations (26)–(27) first give $\|y\|_{L^2} \leq C\eta_t \sqrt{\lambda}$.

To obtain the same estimate in stronger norms, observe that the nonzero β_{ij} form a fixed finite collection in distinct orthogonal blocks. Therefore $\|B_t x\|_{H^s} \leq C_s \|B_t x\|_{L^2}$ for $x \in \mathcal{K}$, uniformly in its scalar coefficients and in t . The coefficients of B_t and all their derivatives are $O(t^{p^*})$, so

$$\|R_t x\|_{H^s} \leq C_s t^{p^*} A_t(x)^{1/2}.$$

Uniform elliptic estimates, together with the L^2 inverse bound, give uniform $H^s \rightarrow H^{s+2}$ bounds for $(T_t - \lambda)^{-1}$ when $0 \leq \lambda \leq \gamma/16$. The projections P, Q introduce only smooth finite-rank terms. Apply these estimates to (27) and then use Sobolev embedding to obtain (30).

The largest small eigenvalue is $O(t^{2p^*})$. Applying (30) to an orthonormal eigenbasis bounds Q on $\text{ran } \Pi_t$, as a map to C^k , by $O(t^{2p^*})$. The spaces $\text{ran } \Pi_t$ and $\mathcal{K}$ have the same finite dimension. Thus P restricts to an isomorphism between them for small t , and $\text{ran } \Pi_t$ is the

graph of a map $\mathcal{K} \rightarrow QH$ of C^k norm $O(t^{2p*})$. The formula for the orthogonal projection onto this graph proves (31). $\square$

After adjoining the scalar identity, the limiting subspace is the algebra $\bigoplus_i \mathbb{C} \text{Id}_{F_i}$. Its minimal nonzero idempotents are the projections onto the F_i , and hence determine the unordered summands of the split limit E_0 . The small spectral subspace at positive t need not be an algebra.

3.3. Mean curvature of the family. The common Einstein constant of the h_i is κ . The Chern connection is $\nabla_t = \nabla_0 + N_t - N_t^*$. Harmonicity and the Kähler identities eliminate the linear term in its mean curvature, so

$$(32) \quad K_t = \kappa \text{Id} + \sqrt{-1} \Lambda_\omega (N_t - N_t^*) \wedge (N_t - N_t^*), \quad K_t - \kappa \text{Id} = O_{C^k}(t^{2p*}).$$

In a unitary coframe on the curve, write $N_t = n_t \bar{\theta}$. The quadratic term is $[n_t, n_t^*]$, so it is trace-free; it may have off-diagonal blocks. Thus $g_t^* h_0$ is an approximate Hermitian–Einstein family on the fixed bundle E_1 . The upper-triangular filtration, whose quotients are stable of the same slope, proves semistability by degree additivity. Its first summand excludes stability. Theorem 3.1 supplies the precise spectral rates for this family.

4. DECAY ORDERS AND LIMITING SPECTRAL SUBSPACES

The graph comparison reduces every leading decay order to a finite construction. Let $p^{(1)} < \dots < p^{(s)}$ be the distinct exponents p_{ij} on nonzero edges. For such an exponent p , let $\mathcal{C}_{< p}$ be the connected components of the graph that retains only edges with $p_{ij} < p$, and define $\mathcal{C}_{\leq p}$ similarly. Isolated vertices count as components. Define a quotient graph on the vertices $A \in \mathcal{C}_{< p}$ by the masses and conductances

$$(33) \quad M_A = \sum_{i \in A} M_i, \quad c_{AB}^{(p)} = \sum_{\substack{ij \text{ joins } A \text{ to } B \\ p_{ij} = p}} c_{ij}, \quad A \neq B.$$

Each original unordered edge is counted once. Internal edges make no contribution. Let $L^{(p)}$ be the weighted Laplacian of this quotient graph, and list its positive eigenvalues as $\rho_{p,1}, \dots, \rho_{p,d_p}$, counting multiplicity. Their number is

$$d_p = |\mathcal{C}_{< p}| - |\mathcal{C}_{\leq p}|.$$

The quotient vertex space embeds isometrically in the original vertex space as the vectors constant on each $A \in \mathcal{C}_{< p}$.

Theorem 4.1. *For each p , exactly d_p eigenvalues of Δ_t have the asymptotics*

$$(34) \quad \lambda(t) = t^{2p}(\rho_{p,k} + o(1)), \quad 1 \leq k \leq d_p.$$

Together with the $c-1$ identically zero eigenvalues, these account for all $m-1$ small eigenvalues. For each fixed positive eigenvalue ρ of $L^{(p)}$, the corresponding spectral subspace converges to the ρ -eigenspace, embedded as block-scalar endomorphisms constant on the components in $\mathcal{C}_{< p}$. More precisely, the orthogonal projections converge in $L^2 \rightarrow C^k$ norm for every k . For a repeated coefficient the assertion concerns the entire subspace.

Proof. First work on the full graph vertex space, including its scalar zero mode, with the mass inner product. Let W_p consist of the vectors constant on $\mathcal{C}_{< p}$. All terms with exponent less than p vanish on W_p and, by self-adjointness, have zero cross blocks relative to $W_p \oplus W_p^\perp$. On $W_p^\perp$, their sum divided by t^{2p} tends uniformly to positive infinity on the unit sphere.

Indeed, if stronger edges exist and $p_- < p$ is their largest exponent, the graph gaps inside their connected components give a lower bound $c't^{2(p-p_-)}$ on that complement.

After division by t^{2p} , the terms with exponent at least p have bounded cross blocks. Their restriction to W_p converges to $L^{(p)}$, with masses exactly as in (33). For a bounded spectral parameter, elimination of $W_p^\perp$ introduces an error tending to zero, because the inverse of its diagonal block tends to zero. The finite-dimensional min–max principle then shows that the $\dim W_p$ bounded eigenvalues of $t^{-2p}L_t$ converge to those of $L^{(p)}$; all remaining eigenvalues tend to infinity. If no stronger edge exists, this is ordinary matrix convergence, with no complementary block.

The positive quotient eigenvalues account for exactly d_p modes at this scale. The component counts telescope to $m - c$ over all exponents, leaving c exact graph zero modes. Remove the scalar mode and apply the relative estimate (21) to obtain (34) for Δ_t .

Now let $u_t = x_t + y_t$ be a unit eigensection with $\lambda(t)/t^{2p} \rightarrow \rho > 0$. By (25), $A_t(x_t) = O(t^{2p})$. Every stronger edge therefore satisfies

$$|(x_t)_i - (x_t)_j| = O(t^{p-p_{ij}}).$$

Every limit of x_t lies in $W_p \cap \mathcal{K}$, and $y_t \rightarrow 0$ in every C^k norm by (30). Test the second equation in (27) against a fixed $z \in W_p \cap \mathcal{K}$. Its energy is $O(t^{2p})$, so (29) bounds the correction by $O(t^{2p+2p_*})$. After division by t^{2p} this correction vanishes. The stronger graph terms vanish on z , the terms with exponent greater than p vanish in the limit, and the remaining terms give

$$L^{(p)}x_0 = \rho x_0.$$

This equality on the full quotient space follows from the tested identity on its mean-zero subspace, since both sides have mean zero.

Choose a small interval about ρ containing no other quotient eigenvalue and not containing zero. The eigenvalue count above gives the dimension of the spectral subspace for which λ/t^{2p} lies in this interval. Limits of orthonormal eigensections remain orthonormal, because their Q components tend to zero. They therefore span the entire ρ -eigenspace. Compactness in the fixed finite-dimensional space $\mathcal{K}$, together with (30), gives convergence of the corresponding orthogonal projections in every $L^2 \rightarrow C^k$ norm. $\square$

5. EXAMPLES WITH UNEQUAL DECAY ORDERS

5.1. Three factors. Take $m = 3$, $\ell = (p + q, q, 0)$, and integers $0 < p < q$. Assume $\beta_{12}, \beta_{23} \neq 0$; the entry β_{13} may vanish. Set

$$A = \|\beta_{12}\|_{L^2}^2, \quad B = \|\beta_{23}\|_{L^2}^2, \quad C = \|\beta_{13}\|_{L^2}^2.$$

Thus $k_{12} = At^{2p}$, $k_{23} = Bt^{2q}$, and $k_{13} = Ct^{2(p+q)}$. The two positive graph eigenvalues are the roots of $\nu^2 - T\nu + J = 0$, where

$$(35) \quad T = k_{12}(M_1^{-1} + M_2^{-1}) + k_{23}(M_2^{-1} + M_3^{-1}) + k_{13}(M_1^{-1} + M_3^{-1}),$$

$$(36) \quad J = \frac{M_1 + M_2 + M_3}{M_1 M_2 M_3} (k_{12}k_{23} + k_{12}k_{13} + k_{23}k_{13}).$$

These identities are the trace and sum of principal two-by-two minors of the mass-normalized graph Laplacian. With $\sigma = 2 \min\{p, q - p\} > 0$, Theorem 3.1 gives

$$(37) \quad \lambda_1(t) = B \left(\frac{1}{M_1 + M_2} + \frac{1}{M_3} \right) t^{2q} (1 + O(t^\sigma)),$$

$$(38) \quad \lambda_2(t) = A \left(\frac{1}{M_1} + \frac{1}{M_2} \right) t^{2p} (1 + O(t^\sigma)), \quad \lambda_3(t) \geq \gamma/4.$$

To check the error terms directly, put $\eta = t^{2(q-p)}$ and $\vartheta = t^{2p}$. The leading terms of T, J have relative errors $O(\eta)$ and $O(\vartheta)$ respectively, and $J/T^2 = O(\eta)$. The larger root is $T(1 + O(\eta))$ and the smaller is $(J/T)(1 + O(\eta))$. The operator comparison adds $O(\vartheta)$. At the t^{2q} scale, the stronger 12 edge has combined vertices 1, 2, explaining the mass $M_1 + M_2$ in (37).

Write $r_{12} = r_1 + r_2$ and $r = r_{12} + r_3$. The limiting eigensections are the normalized versions of

$$\begin{aligned} q_{\text{slow}} &= \text{diag} \left(\frac{r_3}{r} \text{Id}_{F_1}, \frac{r_3}{r} \text{Id}_{F_2}, -\frac{r_{12}}{r} \text{Id}_{F_3} \right), & \|q_{\text{slow}}\|^2 &= \frac{V r_{12} r_3}{r}, \\ q_{\text{fast}} &= \text{diag} \left(\frac{r_2}{r_{12}} \text{Id}_{F_1}, -\frac{r_1}{r_{12}} \text{Id}_{F_2}, 0 \right), & \|q_{\text{fast}}\|^2 &= \frac{V r_1 r_2}{r_{12}}. \end{aligned}$$

Theorem 4.1 gives smooth convergence after fixing the phases by positive inner products with these respective limits. The stronger extension separates the internal F_1, F_2 mode at the larger eigenvalue; the weaker extension separates their combined block from F_3 at the smaller eigenvalue.

For three line bundles, $V = 1$, $A = B = 1$, and $(p, q) = (1, 2)$, the formulas become

$$(39) \quad \lambda_1(t) = \frac{3}{2} t^4 (1 + O(t^2)), \quad \lambda_2(t) = 2 t^2 (1 + O(t^2)).$$

These choices are realized geometrically on a curve of genus at least two. Choose three pairwise nonisomorphic degree-zero line bundles. Every nontrivial degree-zero Hom line bundle has $h^1 = g - 1 > 0$, by the Riemann–Roch argument in Corollary 15.9. Choose harmonic forms of unit L^2 norm for the consecutive edges and set $\beta_{13} = 0$. The resulting bundle is simple and strictly semistable.

5.2. Five factors and a repeated coefficient. Take five pairwise nonisomorphic degree-zero line bundles on such a curve, with $V = 1$ and $\ell = (7, 6, 3, 1, 0)$. Choose unit-norm forms on the path edges 12, 23, 34, 45, whose exponents are 1, 3, 2, 1. One may also include a form of norm 2 on edge 15, of exponent 7. All remaining entries vanish. The quotient graphs have the following positive spectra:

p	components before adding edges of exponent p	positive coefficients
1	$\{1\}, \{2\}, \{3\}, \{4\}, \{5\}$	2, 2
2	$\{1, 2\}, \{3\}, \{4, 5\}$	3/2
3	$\{1, 2\}, \{3, 4, 5\}$	5/6
7 (if present)	$\{1, 2, 3, 4, 5\}$	none

At exponent 2, edge 34 joins masses 1, 2, giving $1 + 1/2$. At exponent 3, edge 23 joins masses 2, 3, giving $1/2 + 1/3$. The optional edge of exponent 7 is internal to a component already connected by stronger edges. Consequently

$$(40) \quad \lambda_1(t) \sim \frac{5}{6} t^6, \quad \lambda_2(t) \sim \frac{3}{2} t^4, \quad \lambda_3(t) \sim 2 t^2, \quad \lambda_4(t) \sim 2 t^2.$$

For the last two eigenvalues, the limiting two-dimensional subspace is spanned by the internal differences on $F_1 \oplus F_2$ and $F_4 \oplus F_5$. Theorem 4.1 specifies this subspace, without choosing individual eigenvectors within it.

6. A HARMONIC NORMAL FORM ON CURVES

Every filtered bundle on a curve can be put in a form in which all extension entries are harmonic. The normal form follows from Dolbeault Hodge theory and a finite sequence of upper-triangular gauge transformations. The condition $\bar{\partial}_0^* \beta = 0$ is used in the scaled-extension construction of Sektnan–Tipler [ST24, Section 5.1, equation (28)]; on a curve it makes every extension entry harmonic. We give the triangular construction explicitly to account for the interactions between successive extensions.

Let X be a compact connected Riemann surface with a fixed Kähler form. Suppose a holomorphic bundle E has a filtration by holomorphic subbundles

$$(41) \quad 0 = E^{(0)} \subset E^{(1)} \subset \dots \subset E^{(m)} = E, \quad F_i = E^{(i)} / E^{(i-1)}.$$

Put $G = \bigoplus_i F_i$, let $\bar{\partial}_0$ be its split Dolbeault operator, and fix Hermitian metrics on the F_i . Harmonicity of a $\text{Hom}(F_j, F_i)$ -valued form always refers to these metrics and the fixed Kähler form.

Lemma 6.1 (Harmonic upper-triangular form). *There is a smooth isomorphism $\Phi : G \rightarrow E$ preserving the filtration and inducing the identity on each quotient F_i , such that*

$$(42) \quad \Phi^{-1} \bar{\partial}_E \Phi = \bar{\partial}_0 + \beta, \quad \beta = \sum_{i < j} \beta_{ij}, \quad \bar{\partial}_0 \beta_{ij} = \bar{\partial}_0^* \beta_{ij} = 0.$$

No stability hypothesis or smallness assumption on the original extension entries is needed for this normal form.

Proof. Choose a smooth splitting $\sigma : G \rightarrow E$ of the filtration, inducing the identity on its quotients. The pulled-back operator has the form $\sigma^{-1} \bar{\partial}_E \sigma = \bar{\partial}_0 + A$, where A is strictly upper triangular. For $\ell \geq 1$, define the subbundle

$$\mathfrak{n}^\ell = \bigoplus_{j-i \geq \ell} \text{Hom}(F_j, F_i) \subset \text{End } G.$$

Then $\mathfrak{n}^a \mathfrak{n}^b \subset \mathfrak{n}^{a+b}$, $\mathfrak{n}^m = 0$, and $\bar{\partial}_0$ preserves these subbundles.

Suppose that the entries of a current upper-triangular form A with $j - i < \ell$ have already been made harmonic. Every smooth $\text{Hom}(F_j, F_i)$ -valued $(0, 1)$ -form on a curve has a Hodge decomposition

$$(43) \quad A_{ij} = H_{ij} A_{ij} + \bar{\partial}_0 v_{ij},$$

where H_{ij} is harmonic projection and v_{ij} is a smooth section. Indeed, the $(0, 2)$ -term in the Dolbeault Hodge decomposition vanishes in complex dimension one. Smoothness follows from elliptic regularity. Choose $u \in \Gamma(\mathfrak{n}^\ell)$ with entries $u_{ij} = -v_{ij}$ when $j - i = \ell$ and zero elsewhere, and set $U_\ell = \text{Id} + u$. This is a globally invertible smooth transformation: its inverse is the finite sum $\text{Id} - u + u^2 - \dots$.

The transformed extension form is exactly

$$A' = U_\ell^{-1} A U_\ell + U_\ell^{-1} \bar{\partial}_0 U_\ell.$$

Since $A \in \Omega^{0,1}(\mathfrak{n}^1)$ and $u \in \Gamma(\mathfrak{n}^\ell)$, the multiplication rule gives

$$(44) \quad A' \equiv A + \bar{\partial}_0 u \pmod{\Omega^{0,1}(\mathfrak{n}^{\ell+1})}.$$

For the derivative term, we use $2\ell \geq \ell + 1$. Thus entries with $j - i < \ell$ are unchanged, and each entry with $j - i = \ell$ becomes $H_{ij}A_{ij}$. Changes to higher entries are retained for the subsequent steps.

Starting with $\ell = 1$ and ending with $\ell = m - 1$ makes every entry harmonic. If $U = U_1 \cdots U_{m-1}$ is the product of the successive transformations, then $\Phi = \sigma U$ has all the asserted properties. Each transformation is the identity on the quotients, and the process terminates after finitely many steps. $\square$

The higher entries in (42) need not be the harmonic projections of the original entries of A . For three blocks, let the first correction have entries u_{12}, u_{23} and zero 13 entry, so that $\beta_{12} = A_{12} + \bar{\partial}_0 u_{12}$ and $\beta_{23} = A_{23} + \bar{\partial}_0 u_{23}$ are harmonic. Before the second correction, the 13 entry is

$$(45) \quad A'_{13} = A_{13} + A_{12}u_{23} - u_{12}\beta_{23}.$$

It is this modified entry whose harmonic part defines β_{13} . The induction therefore includes the nonlinear interaction of the successive extensions.

7. THE FLOW FOR TWO FACTORS: THE MODEL CASE

This section treats a nonsplit extension of two stable bundles directly. It introduces the three tools used in general in Sections 8–10: comparison of metrics along the flow, a corrected approximate solution with integrable error, and matching of the flow with the extension orbit at late times.

Let X be a compact connected Riemann surface, and retain F, G , the nonzero harmonic extension form β , and the notation q, V_q, c, C, z, R of Section 2. Thus $E = E_1$ is the nonsplit extension of the nonisomorphic stable equal-slope bundles G by F . We use flow time τ to distinguish it from the scaling parameter of the earlier sections. Starting from any smooth Hermitian metric on E , consider the normalized Hermitian Yang–Mills flow

$$(46) \quad h^{-1}\partial_\tau h = -2(K_h - \kappa \text{Id}), \quad \kappa = \frac{2\pi\mu(E)}{V}.$$

The factor 2 fixes the time normalization in the result below.

We use the classical convergence theorem for Yang–Mills flow on a curve: the solution exists for all time, and in unitary gauges its Chern connections converge to the Hermitian–Einstein connection on the graded bundle $F \oplus G$ [Das92, Råd92]. For smooth initial data the convergence, modulo gauge, is smooth; the higher regularity follows from parabolic regularity. In metric language there are holomorphic isometries from $(E, h(\tau))$ to $(F \oplus G, h_0, \bar{\partial}^{(\tau)})$ with $\bar{\partial}^{(\tau)} \rightarrow \bar{\partial}_0$ smoothly. This analytic convergence and the identification of the graded limit are inputs here.

The metric asymptotics of this flow have been studied much more generally by Haiden–Katzarkov–Kontsevich–Pandit [HKKP18]. Their construction uses harmonic extension forms, a Green operator correction, and the order-preserving property of the metric flow. We use these ingredients in the two-factor case, together with a comparison begun at arbitrarily late times, to determine the leading coefficient of the small Dolbeault eigenvalue.

Theorem 7.1. *For every smooth initial metric in (46), exactly one eigenvalue on trace-free endomorphisms tends to zero, and*

$$(47) \quad \lambda_1(h(\tau)) = \frac{1 + o(1)}{2\tau}, \quad \lambda_2(h(\tau)) \geq \delta > 0$$

for sufficiently large τ .

In the unitary identifications supplied by smooth convergence of the connections, the first spectral projection converges in every $L^2(h_0) \rightarrow C^k(h_0)$ norm to the projection onto $\mathbb{C}q$. Equivalently, the unit first eigensection, with its phase fixed by positive inner product with q , tends smoothly to $q/\sqrt{V_q}$. Its limit recovers the two summand projections by

$$P_F = \sqrt{V_q} u_\infty + \frac{a}{r} \text{Id}, \quad P_G = \text{Id} - P_F.$$

The proof separates metric comparison, a corrected approximate solution, and a local description of the fixed extension orbit.

7.1. Order comparison and the endomorphism spectrum. If two metrics h_1, h_2 on a fixed holomorphic bundle satisfy $e^{-\epsilon} h_1 \leq h_2 \leq e^\epsilon h_1$, then their induced norms on endomorphisms satisfy

$$e^{-2\epsilon} |u|_{h_1}^2 \leq |u|_{h_2}^2 \leq e^{2\epsilon} |u|_{h_1}^2.$$

Indeed, in an h_1 -orthonormal frame diagonalizing h_2 , the coefficient of $|u_{ij}|^2$ is the ratio of its i th and j th eigenvalues. The same bounds hold for endomorphism-valued $(0, 1)$ -forms, since the base metric is fixed. The Dolbeault operator and its trace-free domain are independent of the bundle metric, so min–max gives

$$(48) \quad e^{-4\epsilon} \lambda_j(h_1) \leq \lambda_j(h_2) \leq e^{4\epsilon} \lambda_j(h_1).$$

A positive scalar multiple of either metric leaves these eigenvalues unchanged.

The order comparison for the metric flow has a useful version with an error term. Suppose a smooth family $\hat{h}(\tau)$ has residual

$$(49) \quad \mathcal{E}_{\hat{h}} := \hat{h}^{-1} \partial_\tau \hat{h} + 2(K_{\hat{h}} - \kappa \text{Id}), \quad \|\mathcal{E}_{\hat{h}}\|_{L^\infty, \text{op}, \hat{h}} \leq f(\tau).$$

If h is an exact solution and $e^{-\epsilon} \hat{h}(T) \leq h(T) \leq e^\epsilon \hat{h}(T)$, then

$$(50) \quad e^{-\epsilon - \int_T^\tau f(s) ds} \hat{h}(\tau) \leq h(\tau) \leq e^{\epsilon + \int_T^\tau f(s) ds} \hat{h}(\tau).$$

To see this, multiply $\hat{h}$ by $e^{\pm \epsilon \pm \int_T^\tau f}$. A spatially constant scalar does not change Chern curvature, and its logarithmic derivative adds $\pm f \text{Id}$ to the residual. The two metrics are therefore respectively a subsolution and a supersolution. The maximum principle for Hermitian metrics gives the assertion; this is the order comparison of [HKKP18, Section 4.2, Propositions 4.3 and 4.5], with the error measured in the normalization (49).

7.2. An approximate solution with a small total error. Use determinant-one diagonal scaling

$$(51) \quad g_\rho = \rho^q = \text{diag}(\rho^{b/r} \text{Id}_F, \rho^{-a/r} \text{Id}_G), \quad h_\rho = g_\rho^* h_0.$$

It still transports the extension entry to $\rho\beta$, and its endomorphism spectrum is that of Theorem 2.1. The uncorrected curvature is $\kappa \text{Id} + \rho^2 C$. Its component $\rho^2 z$ perpendicular to q generally prevents h_ρ from following (46) under a scalar change of parameter.

Put

$$(52) \quad v = Rz, \quad k_\rho = \exp(-\tfrac{1}{2}\rho^2 v), \quad \hat{g}_\rho = k_\rho g_\rho, \quad \hat{h}_\rho = \hat{g}_\rho^* h_0.$$

The endomorphism v is smooth, self-adjoint, trace-free, and block diagonal, so it commutes with q . Block diagonality follows because the split Laplacian preserves the diagonal blocks. Self-adjointness follows from the Kähler identities: at a Hermitian–Einstein connection the endomorphism Dolbeault Laplacian is half the connection Laplacian and therefore preserves adjoints.

Lemma 7.2. *Let $0 < \rho_0 \leq \rho_*$ for a fixed sufficiently small ρ_* , and define, for $\tau \geq T$,*

$$(53) \quad \rho(\tau) = (\rho_0^{-2} + 2c(\tau - T))^{-1/2}.$$

There is a constant A , independent of T and ρ_0 , such that $\widehat{h}(\tau) = \widehat{h}_{\rho(\tau)}$ satisfies

$$(54) \quad \|\mathcal{E}_{\widehat{h}}\|_{L^\infty, \text{op}, \widehat{h}} \leq A\rho(\tau)^3, \quad \int_T^\infty A\rho(\tau)^3 d\tau = \frac{A}{c}\rho_0.$$

For this approximate solution,

$$(55) \quad \lambda_1(\widehat{h}_\rho) = c\rho^2(1 + O(\rho^2)), \quad \lim_{\tau \rightarrow \infty} \tau \lambda_1(\widehat{h}(\tau)) = \frac{1}{2}.$$

Proof. Transport everything by $\widehat{g}_\rho$ to the fixed metric h_0 . At the split structure the derivative of mean curvature under the metric change $h_0 e^{\epsilon s}$ is $\Delta_0 s$. For example, this follows by differentiating the Chern curvature to obtain $\sqrt{-1}\Lambda \bar{\partial}_0 \partial_0 s = \Delta_0 s$. Smooth Taylor expansion in ρ , with $k_\rho = \text{Id} - \frac{1}{2}\rho^2 v + O_{C^k}(\rho^4)$, therefore gives

$$(56) \quad \widehat{K}_\rho := \widehat{g}_\rho K_{\widehat{h}_\rho} \widehat{g}_\rho^{-1} = \kappa \text{Id} + \rho^2(C - \Delta_0 v) + O_{C^k}(\rho^3) = \kappa \text{Id} + c\rho^2 q + O_{C^k}(\rho^3).$$

Terms involving both the change of metric, of order ρ^2 , and the extension, of order ρ , are included in the cubic remainder. All estimates are uniform for $0 < \rho \leq \rho_*$.

Since q and v commute and $\dot{\rho} = -c\rho^3$, the logarithmic derivative of the gauge is exactly

$$\dot{\widehat{g}}_\rho \widehat{g}_\rho^{-1} = -c\rho^2 q + c\rho^4 v.$$

For a metric $g^* h_0$, its logarithmic time derivative, transported by g , is $\dot{g}g^{-1} + (\dot{g}g^{-1})^*$. Combining this with (56) cancels the quadratic terms in (49) and proves the cubic bound. Integration using $\dot{\rho} = -c\rho^3$ gives the stated total error.

Also $e^{-B\rho^2} h_\rho \leq \widehat{h}_\rho \leq e^{B\rho^2} h_\rho$ for some fixed B . Apply (48) and Theorem 2.1 to get $\lambda_1(\widehat{h}_\rho) = c\rho^2(1 + O(\rho^2))$. Equation (53) proves the limit. $\square$

7.3. Matching the actual flow at late times. The following observation strengthens bounded comparison to comparison with a factor tending to one. It uses the fact that the extension has two nonisomorphic stable factors.

Lemma 7.3. *For the solution $h(T)$ of (46), there are $\rho_T > 0$, $A_T > 0$, and $\epsilon_T \geq 0$, with $\rho_T \rightarrow 0$ and $\epsilon_T \rightarrow 0$, such that*

$$(57) \quad e^{-\epsilon_T} A_T \widehat{h}_{\rho_T} \leq h(T) \leq e^{\epsilon_T} A_T \widehat{h}_{\rho_T}.$$

Proof. Use the smooth convergence of the flow in unitary gauges, and write $D_T = \bar{\partial}^{(T)} \rightarrow \bar{\partial}_0$ on $(F \oplus G, h_0)$, with each $(F \oplus G, D_T)$ holomorphically isomorphic to E . The spaces of holomorphic maps $F \rightarrow E$ and $E \rightarrow G$ both have dimension one: apply $\text{Hom}(F, \cdot)$ and $\text{Hom}(\cdot, G)$ to the extension and use stability and $\text{Hom}(F, G) = 0$. Their counterparts at $F \oplus G$ are also one-dimensional, generated by the inclusion and projection.

The Dolbeault Laplacians on these two Hom bundles converge smoothly. The spectral projections onto their one-dimensional kernels therefore converge in every smooth norm, by elliptic resolvent estimates. Choose holomorphic maps $i_T : F \rightarrow (F \oplus G, D_T)$ and $p_T : (F \oplus G, D_T) \rightarrow G$ converging respectively to inclusion and projection. For large T , i_T is an embedding and p_T is surjective. Moreover $p_T i_T = 0$, since $\text{Hom}(F, G) = 0$.

Set $j_T = p_T^*(p_T p_T^*)^{-1}$ and $\sigma_T(f, g) = i_T f + j_T g$. This smooth splitting tends to the identity and has $p_T \sigma_T(f, g) = g$. Holomorphicity of i_T, p_T implies

$$\sigma_T^{-1} D_T \sigma_T = \begin{pmatrix} \bar{\partial}_F & \alpha_T \\ 0 & \bar{\partial}_G \end{pmatrix}, \quad \alpha_T \rightarrow 0 \quad \text{in } C^\infty.$$

Hodge decomposition writes $\alpha_T = H\alpha_T + \bar{\partial}v_T$, with $v_T \rightarrow 0$ smoothly. The unipotent change of splitting with upper-right entry $-v_T$ makes the extension entry $H\alpha_T$, while the resulting isomorphism S_T still tends to the identity.

The harmonic form $H\alpha_T$ is a nonzero scalar multiple $z_T\beta$. Indeed, an isomorphism with E identifies its inclusion of F and quotient map to G up to nonzero scalars, because the two Hom spaces above have dimension one. It therefore identifies the extension classes up to their scalar ratio. Harmonic representatives preserve this equality. Nonvanishing follows from nonsplitting, and $z_T \rightarrow 0$ follows from $\alpha_T \rightarrow 0$.

Set $\rho_T = |z_T|$. A block-scalar unitary transformation removes the phase of z_T . The pulled-back metric on $(F \oplus G, \bar{\partial}_0 + \rho_T N)$ tends to h_0 smoothly, since $S_T \rightarrow \text{Id}$ and the phase transformation is unitary and parallel. Its holomorphic identification with $(E, h(T))$ differs from $g_{\rho_T} : E \rightarrow E_{\rho_T}$ by a scalar automorphism of E , using simplicity from Theorem 2.1. Consequently, for some $A_T > 0$,

$$h(T) = a_T g_{\rho_T}^* \tilde{h}_T, \quad \tilde{h}_T \rightarrow h_0 \quad \text{in } C^\infty.$$

Thus $h(T)$ is within a multiplicative factor tending to one of $a_T h_{\rho_T}$. The same is true of $\hat{h}_{\rho_T}$ relative to h_{ρ_T} by (52), which proves (57). $\square$

Proof of Theorem 7.1. For a sufficiently large T , start the approximate solution of Lemma 7.2 with $\rho_0 = \rho_T$. Multiply it by A_T , which changes neither its curvature nor its endomorphism spectrum. Lemma 7.3 supplies its initial comparison with $h(T)$. Equations (50) and (54) give, for every $\tau \geq T$,

$$e^{-b_T} A_T \hat{h}(\tau) \leq h(\tau) \leq e^{b_T} A_T \hat{h}(\tau), \quad b_T = \epsilon_T + (A/c)\rho_T \rightarrow 0.$$

It follows from (48) and (55) that

$$\frac{1}{2}e^{-4b_T} \leq \liminf_{\tau \rightarrow \infty} \tau \lambda_1(h(\tau)) \leq \limsup_{\tau \rightarrow \infty} \tau \lambda_1(h(\tau)) \leq \frac{1}{2}e^{4b_T}.$$

Now let $T \rightarrow \infty$ to prove (47) for the first eigenvalue. A comparison begun at a single fixed time would give only two-sided bounds of order τ^{-1} , not the coefficient.

In the unitary gauges of smooth connection convergence, the endomorphism Laplacians converge smoothly as operators with domain H^2 . Their limit is Δ_0 , whose kernel on trace-free endomorphisms is $\mathbb{C}q$. A contour enclosing only zero therefore gives spectral projections of rank one for large τ and a uniform gap above them. Resolvent convergence on the Sobolev scale gives smooth convergence of the normalized eigensections and hence of the finite-rank projections in $L^2 \rightarrow C^k$ norm for every k . This proves the remaining claims and the recovery formulas. $\square$

The leading coefficient in (47) is independent of the ranks, extension norm, and initial metric under the normalization (46). These data enter the intermediate constant c , which cancels between the model spectrum $c\rho^2$ and the flow law $\rho^2 \sim (2c\tau)^{-1}$. The theorem does not assert a next-order eigenvalue expansion or a quantitative rate of eigensection convergence along the actual flow. In particular, the fourth-order coefficient in (11) cannot simply be transferred by substituting $\rho = (2c\tau)^{-1/2}$ into the uncorrected harmonic family. Sections 8

to 10 extend the argument to every simple bundle on a curve with pairwise nonisomorphic Jordan–Hölder factors, including eigenvalues with distinct scales.

8. THE DIAGONAL FLOW

This section is finite-dimensional. It determines the exact asymptotics of the flow on block-scalar metrics that governs the small spectrum in Sections 9 and 10.

8.1. Setting and the weight grading. Let Γ be a finite directed acyclic graph with vertex set $\{1, \dots, m\}$ and edge set Γ_1 . An edge $\alpha \in \Gamma_1$ from i to j is written $\alpha = (i \rightarrow j)$. Fix masses $M_i > 0$ and constants $c_\alpha > 0$, and assume that the underlying undirected graph is connected. In the geometric application the vertices are the stable factors, an edge $i \rightarrow j$ records a nonzero harmonic entry β_{ij} with F_i on the subbundle side, $M_i = V \operatorname{rk} F_i$, and $c_\alpha = \|\beta_{ij}\|_{L^2}^2$.

On the hyperplane $\sum_i M_i y_i = 0$ of $\mathbb{R}^m$ put

$$(58) \quad k_\alpha(y) = c_\alpha e^{2(y_i - y_j)}, \quad M_i p_i(y) = \sum_{\alpha \text{ out of } i} k_\alpha - \sum_{\alpha \text{ into } i} k_\alpha, \quad \dot{y}_i = -p_i(y).$$

With $d_i = e^{y_i}$ this is the flow $\dot{g}g^{-1} = -\operatorname{diag}(p_i \operatorname{Id}_{F_i})$ for $g = \operatorname{diag}(d_i \operatorname{Id}_{F_i})$ studied in Section 9. Put $\mathfrak{e} = \sum_\alpha k_\alpha$, the *total conductance*. The positive eigenvalues $\nu_1 \leq \dots \leq \nu_{m-1}$ of the weighted graph form $\sum_\alpha k_\alpha |x_i - x_j|^2$ on $\{\sum_i M_i x_i = 0\}$, with norm $\sum_i M_i |x_i|^2$, are the *graph eigenvalues* of y .

Definition 8.1. The *weight grading* is the unique minimizer $w \in \mathbb{R}^m$ of $\frac{1}{2} \sum_i M_i w_i^2$ subject to $w_j - w_i \geq \frac{1}{2}$ for every edge $i \rightarrow j$. An edge is *tight* if equality holds, and its *slack* is $\varepsilon_\alpha = 2(w_j - w_i) - 1 \geq 0$. The *tight subgraph* $\mathcal{T} \subset \Gamma_1$ consists of the tight edges, and its connected components (on all m vertices, isolated vertices included) are the *tight components*.

The constraint set is nonempty, since half the length of a longest path into each vertex is admissible, and the objective is strictly convex. The constraints are invariant under adding constants, so $\sum_i M_i w_i = 0$. Since they are linear, there are Lagrange multipliers $u_\alpha \geq 0$, vanishing off $\mathcal{T}$, with

$$(59) \quad M_i w_i = \sum_{\alpha \text{ into } i} u_\alpha - \sum_{\alpha \text{ out of } i} u_\alpha.$$

Summing over a tight component $\mathcal{C}$ gives $\sum_{i \in \mathcal{C}} M_i w_i = 0$, because u is supported on edges inside tight components. A vertex with no tight edge therefore has $w_i = 0$. At least one edge is tight: otherwise every multiplier would vanish, so $w = 0$ by (59), which is not admissible.

This is the weight grading of Haiden, Katzarkov, Kontsevich, and Pandit [HKKP23, Section 3]. Their flow [HKKP23, (3.10)] is (58) after the substitutions

$$x = -2y, \quad t = 2\tau, \quad v = -2w, \quad m_i = M_i,$$

and their tight edges $v_i - v_j = 1$ are ours.

Definition 8.2. The pair (Γ, M) is *nondegenerate* if some multiplier vector satisfying (59) is strictly positive on every tight edge.

In the language of optimization this is strict complementarity. Replacing M by a positive multiple leaves w unchanged and rescales the multipliers, so for $M_i = V \operatorname{rk} F_i$ nondegeneracy depends only on Γ and the ranks. We then say that $(\Gamma, (r_i))$ is nondegenerate.

This is exactly the hypothesis of [HKKP23, Theorem 3.6], under which $y = w \log \tau + O(1)$. It fails, for example, for the zigzag $1 \rightarrow 2 \leftarrow 3 \rightarrow 4$ with masses satisfying $M_1 M_4 = M_2 M_3$

(compare [HKKP23, (3.23)–(3.28)]), where iterated logarithms can appear. The equal-rank zigzag of Theorem 11.4 lies on that quadric. In a nondegenerate graph the tight edges are exactly the edges carrying positive multipliers.

8.2. Convergence of the remainder.

Theorem 8.3. *Assume (Γ, M) is nondegenerate. The tight potential*

$$\Phi(z) = \sum_i M_i w_i z_i + \frac{1}{2} \sum_{\alpha \in \mathcal{T}} c_\alpha e^{2(z_i - z_j)}$$

has a minimizer ζ^ , unique subject to $\sum_{i \in \mathcal{C}} M_i \zeta_i^* = 0$ for every tight component $\mathcal{C}$. Every solution of (58) satisfies*

$$(60) \quad y(\tau) = w \log \tau + \zeta^* + \sum_{\mathcal{C}} Z_{\mathcal{C}} \mathbf{1}_{\mathcal{C}} + O(\tau^{-\delta})$$

for some $\delta > 0$ and constants $Z_{\mathcal{C}}$ with $\sum_{\mathcal{C}} M_{\mathcal{C}} Z_{\mathcal{C}} = 0$, where $M_{\mathcal{C}} = \sum_{i \in \mathcal{C}} M_i$ and $\mathbf{1}_{\mathcal{C}}$ is the indicator vector of $\mathcal{C}$. The vector ζ^ depends only on (Γ, M, c) . The constants $Z_{\mathcal{C}}$ depend on the solution.*

Proof. Set $s = \log \tau$ for $\tau \geq 1$ and $z(s) = y(e^s) - ws$. Then $\sum_i M_i z_i = 0$, and $\tau k_\alpha = c_\alpha e^{2t_\alpha} e^{-\varepsilon_\alpha s}$ with $t_\alpha = z_i - z_j$. Define

$$\Psi_s(z) = \sum_i M_i w_i z_i + \frac{1}{2} \sum_{\alpha} c_\alpha e^{2t_\alpha} e^{-\varepsilon_\alpha s}.$$

Since $\partial \Psi_s / \partial z_i = M_i w_i + M_i \tau p_i$ and $dz_i / ds = -\tau p_i - w_i$, the curve z is the gradient flow $dz/ds = -M^{-1} \nabla \Psi_s(z)$. Because $\partial_s \Psi_s \leq 0$,

$$(61) \quad \frac{d}{ds} \Psi_s(z(s)) = - \sum_i \frac{(\partial_i \Psi_s)^2}{M_i} - \frac{1}{2} \sum_{\alpha} \varepsilon_\alpha \tau k_\alpha \leq 0.$$

Fix multipliers u satisfying (59) with $u_\alpha > 0$ on $\mathcal{T}$. By (59), $\sum_i M_i w_i z_i = -\sum_{\alpha} u_\alpha t_\alpha$, so

$$\Psi_s(z) = \sum_{\alpha \in \mathcal{T}} \left(\frac{c_\alpha}{2} e^{2t_\alpha} - u_\alpha t_\alpha \right) + \frac{1}{2} \sum_{\alpha \notin \mathcal{T}} c_\alpha e^{2t_\alpha} e^{-\varepsilon_\alpha s}.$$

Each tight term is bounded below by $\frac{u_\alpha}{2} (1 - \log(u_\alpha/c_\alpha))$ and is proper in t_α ; the other terms are nonnegative. Hence $\Psi_s(z(s))$ decreases to a finite limit, every tight difference t_α is bounded along the solution, and (61) gives

$$(62) \quad \int_0^\infty \sum_{\alpha \notin \mathcal{T}} \varepsilon_\alpha \tau k_\alpha ds < \infty.$$

Within a tight component, $z_i - z_j$ is therefore bounded.

Let $Z_{\mathcal{C}}(s) = M_{\mathcal{C}}^{-1} \sum_{i \in \mathcal{C}} M_i z_i$. In $\sum_{i \in \mathcal{C}} \partial_i \Psi_s$ every edge inside $\mathcal{C}$ cancels, and $\sum_{i \in \mathcal{C}} M_i w_i = 0$. Edges between different tight components are not tight, so their slack is at least $\varepsilon_* = \min_{\alpha \notin \mathcal{T}} \varepsilon_\alpha > 0$. Hence

$$\left| \frac{dZ_{\mathcal{C}}}{ds} \right| \leq \frac{1}{M_{\mathcal{C}} \varepsilon_*} \sum_{\alpha \notin \mathcal{T}} \varepsilon_\alpha \tau k_\alpha,$$

which is integrable by (62). So $Z_{\mathcal{C}}(s)$ converges, and z is bounded. Consequently $\tau k_\alpha \leq C e^{-\varepsilon_* s}$ for $\alpha \notin \mathcal{T}$. If every edge is tight there is one component, and these terms are absent.

Let H be the M -orthogonal complement of the span of the $\mathbf{1}_C$, and write $z = \zeta + \sum_C Z_C \mathbf{1}_C$ with $\zeta \in H$. The tight potential satisfies $\Phi(z + \sum_C \theta_C \mathbf{1}_C) = \Phi(z)$, so $M^{-1} \nabla \Phi$ takes values in H , and

$$\frac{d\zeta}{ds} = -M^{-1} \nabla \Phi(\zeta) + R(s), \quad |R(s)| \leq C e^{-\varepsilon_* s}.$$

On H the map $\zeta \mapsto (t_\alpha)_{\alpha \in \mathcal{T}}$ is injective, so $\Phi|_H$ is strictly convex and proper, with a unique minimizer ζ^* , and its Hessian $2 \sum_{\alpha \in \mathcal{T}} c_\alpha e^{2t_\alpha} (dt_\alpha)^2$ is uniformly positive on bounded subsets of H .

Let $B \subset H$ be a closed ball containing ζ^* and the bounded curve $\zeta([0, \infty))$. On B , in the norm $|x|_M^2 = \sum_i M_i x_i^2$, the Hessian of $\Phi|_H$ is bounded below by some $c_0 > 0$. Put $W = \Phi(\zeta) - \Phi(\zeta^*) \geq 0$. For $\zeta \in B$, strong convexity gives

$$\frac{c_0}{2} |\zeta - \zeta^*|_M^2 \leq W \leq \frac{1}{2c_0} |\nabla \Phi(\zeta)|_{M^{-1}}^2,$$

where $|v|_{M^{-1}}^2 = \sum_i v_i^2 / M_i$. Along the solution,

$$\frac{dW}{ds} = -|\nabla \Phi|_{M^{-1}}^2 + \langle \nabla \Phi, R \rangle \leq -\frac{1}{2} |\nabla \Phi|_{M^{-1}}^2 + \frac{1}{2} |R|_M^2 \leq -c_0 W + C^2 e^{-2\varepsilon_* s}.$$

By Grönwall's inequality, $W(s) \leq C'(1+s)e^{-\min(c_0, 2\varepsilon_*)s}$, and hence $|\zeta(s) - \zeta^*|_M = O(e^{-\delta s})$ for every $\delta < \frac{1}{2} \min(c_0, 2\varepsilon_*)$. Since z is bounded, $|dZ_C/ds| \leq C e^{-\varepsilon_* s}$, and so $|Z_C(s) - \lim Z_C| \leq C e^{-\varepsilon_* s} / \varepsilon_*$. In the variable $\tau = e^s$ this is (60). $\square$

The critical equation of Φ on H states that

$$(63) \quad u_\alpha^* = c_\alpha e^{2(\zeta_i^* - \zeta_j^*)} \quad (\alpha \in \mathcal{T})$$

satisfies (59). It is the unique multiplier vector of this form. When $\mathcal{T}$ is a forest, (59) determines u^* , which is then independent of c .

Remark 8.4. After the substitution $\theta = M^{1/2}y$, the flow (58) is gradient flow for the exponential loss $\frac{1}{2} \sum_\alpha c_\alpha e^{-\langle x_\alpha, \theta \rangle}$ of the linearly separable vectors $x_\alpha = 2M^{-1/2}(e_j - e_i)$. The weight grading is the corresponding hard-margin direction, and tight edges are support vectors. For gradient descent with unit weights, Soudry, Hoffer, Nacson, Gunasekar, and Srebro prove convergence of the remainder when the support vectors span the data [SHNGS18, Theorems 3 and 4]. Here that means there is a single tight component. Nacson et al. [NLGSSS19, Theorem 10] bound the component of the remainder outside the span of the support vectors without that assumption. The iterated-logarithm expansion of Soudry et al. in the degenerate case [SHNGS18, Theorem 13] parallels [HKKP23, Section 5]. The case of several tight components, where the constants Z_C carry the dependence on the solution, is the case needed below. The monotone quantity Ψ_s handles it directly.

8.3. Conductances and graph eigenvalues. Put $a_\alpha = 1 + \varepsilon_\alpha$ and $u_\alpha^0 = c_\alpha e^{2(\zeta_i^* - \zeta_j^*)}$ for every edge; on tight edges $u_\alpha^0 = u_\alpha^*$. By Theorem 8.3,

$$(64) \quad \tau^{a_\alpha} k_\alpha(y(\tau)) \longrightarrow k_\alpha^\infty = u_\alpha^0 e^{2(Z_{\mathcal{C}(i)} - Z_{\mathcal{C}(j)})},$$

where $\mathcal{C}(i)$ is the tight component of i . In particular $\tau k_\alpha \rightarrow u_\alpha^*$ on tight edges, whatever the solution.

For an exponent a , contract the connected components of the subgraph of edges with $a_\alpha < a$, adding masses. Join the resulting vertices by the edges with $a_\alpha = a$ that connect distinct contracted vertices, with conductances k_α^∞ summed over parallel edges. Let d_a be the number of positive eigenvalues of this a -quotient graph. Edges with $a_\alpha = a$ inside one contracted vertex are omitted; such edges can occur.

Theorem 8.5. *Assume (Γ, M) is nondegenerate, and let y solve (58).*

- (1) *For each exponent a , exactly d_a graph eigenvalues satisfy $\nu = \rho \tau^{-a}(1 + o(1))$, where ρ runs through the positive eigenvalues of the a -quotient graph, with multiplicity. The corresponding spectral subspaces converge to the quotient eigenspaces, embedded as vectors constant on the contracted vertices.*
- (2) *At $a = 1$ the quotient graph is the tight subgraph, which is nonempty, with conductances u^* and masses M_i . These coefficients are the same for every solution. The value $\frac{1}{2}$ occurs with multiplicity at least the number of tight components with at least two vertices. For each such component $\mathcal{C}$, $w\mathbf{1}_{\mathcal{C}}$ is a $\frac{1}{2}$ -eigenvector.*
- (3) *For each $a > 1$ with $d_a \geq 1$, the sum of the coefficients at exponent a is*

$$\sum_{\alpha} k_{\alpha}^{\infty} (M_A^{-1} + M_B^{-1}),$$

summed over edges with $a_{\alpha} = a$ that join distinct contracted vertices $A \neq B$. As the initial condition varies, this sum is unbounded, and so is the largest coefficient at exponent a .

Proof. By (64), each conductance lies within a factor $1 + O(\tau^{-\delta})$ of $k_{\alpha}^{\infty} \tau^{-a_{\alpha}}$. A common relative perturbation of all conductances moves every graph eigenvalue by at most the same factor, by min–max. The proof of Theorem 4.1 uses only the ordering of the exponents. With $t = \tau^{-1}$ and real exponents a_{α} in place of $2p_{ij}$, it gives the eigenvalue and eigenspace statements in (1); the $1 + o(1)$ factors change each quotient by $o(1)$.

In (2) nothing is contracted at $a = 1$, and (63) identifies the weights. For every tight edge, $w_j - w_i = \frac{1}{2}$. By (59),

$$\sum_{\alpha \in \mathcal{T}, \alpha \ni i} u_{\alpha}^* (w_i - w_{\alpha(i)}) = \frac{1}{2} \left(\sum_{\alpha \text{ into } i} u_{\alpha}^* - \sum_{\alpha \text{ out of } i} u_{\alpha}^* \right) = \frac{1}{2} M_i w_i,$$

where $\alpha(i)$ is the other endpoint. This identity holds row by row, and the tight Laplacian is block diagonal over tight components. So $w\mathbf{1}_{\mathcal{C}}$ is a $\frac{1}{2}$ -eigenvector whenever it is nonzero. It is nonzero when $\mathcal{C}$ has an edge, and it is M -orthogonal to $\mathbf{1}_{\mathcal{C}}$.

For (3), the trace of the Laplacian of the a -quotient graph with respect to the masses is the displayed sum. Its terms are $u_{\alpha}^0 e^{2(Z_{\mathcal{C}(i)} - Z_{\mathcal{C}(j)})}$ with $\mathcal{C}(i) \neq \mathcal{C}(j)$, and the linear forms $Z_{\mathcal{C}(i)} - Z_{\mathcal{C}(j)}$ are nonzero on $\{\sum M_{\mathcal{C}} Z_{\mathcal{C}} = 0\}$. So the sum is unbounded as Z ranges over this hyperplane.

Every such Z is approximately attained. Start at a late time s_0 from $\zeta^* + \sum_{\mathcal{C}} \theta_{\mathcal{C}} \mathbf{1}_{\mathcal{C}}$, with $\sum_{\mathcal{C}} M_{\mathcal{C}} \theta_{\mathcal{C}} = 0$, and put $\Theta = \max |\theta_{\mathcal{C}}|$. Let $s_2 \in (s_0, \infty]$ be the supremum of the s for which $|z(\sigma) - z(s_0)|_{\infty} \leq 1$ on $[s_0, s]$. On $[s_0, s_2]$, every difference t_{α} is at most $2\Theta + 2 + \max |\zeta_i^* - \zeta_j^*|$. So every non-tight conductance satisfies $\tau k_{\alpha} \leq C_1 e^{4\Theta} e^{-\varepsilon_* s}$, where C_1 does not depend on $(\theta_{\mathcal{C}})$ or s_0 . Let B now be the closed ball in H about ζ^* containing all ζ with $|z - z(s_0)|_{\infty} \leq 1$, for $z(s_0)$ of the above form. Its convexity constant c_0 depends only on Γ , M , c and Θ . There:

- $|dZ_{\mathcal{C}}/ds| \leq C_2 e^{4\Theta} e^{-\varepsilon_* s}$, so $|Z_{\mathcal{C}}(s) - \theta_{\mathcal{C}}| \leq C_2 e^{4\Theta - \varepsilon_* s_0} / \varepsilon_*$;
- the transverse part starts at the critical point, $\zeta(s_0) = \zeta^*$. The estimate above, with $R = O(e^{4\Theta - \varepsilon_* s})$, gives $|\zeta(s) - \zeta^*|_M \leq C_3 e^{4\Theta - \varepsilon_* s_0}$.

The deviation $|z(s) - z(s_0)|_{\infty}$ is at most $C_2 e^{4\Theta - \varepsilon_* s_0} / \varepsilon_* + (\min_i M_i)^{-1/2} C_3 e^{4\Theta - \varepsilon_* s_0}$. If s_0 is so large that this is below $\frac{1}{2}$, the deviation stays below $\frac{1}{2}$ on $[s_0, s_2]$. By continuity this forces $s_2 = \infty$. Hence

$$\lim_{s \rightarrow \infty} Z_{\mathcal{C}}(s) = \theta_{\mathcal{C}} + O(e^{4\Theta - \varepsilon_* s_0}).$$

The flow is autonomous, and a shift of time does not change the leading coefficient of τ^{-a} . So every vector (θ_C) is, up to an arbitrarily small error, the vector of limits Z_C of some solution. $\square$

9. UNIFORM SPECTRAL TRANSFER

Throughout this section X is a compact curve. Let E be a simple semistable holomorphic bundle whose Jordan–Hölder factors $F_1, \dots, F_m$, $m \geq 2$, are pairwise nonisomorphic, with ranks r_i and Hermitian–Einstein metrics h_i . Put $E_0 = \bigoplus_i F_i$, $h_0 = \bigoplus_i h_i$, $M_i = Vr_i$, and $|M| = \sum_i M_i$.

9.1. The socle filtration and the harmonic normal form. The *socle* $\text{soc}(E)$ is the sum of the stable subbundles of E of slope $\mu(E)$; it is polystable. Put $S_1 = \text{soc}(E)$ and let $S_{k+1} \supset S_k$ be the preimage of $\text{soc}(E/S_k)$. This *socle filtration* $0 = S_0 \subset S_1 \subset \dots \subset S_L = E$ is canonical, and each S_k/S_{k-1} is a direct sum of some of the F_i . Since the factors are pairwise nonisomorphic, each F_i occurs in exactly one S_k/S_{k-1} ; we call k the *socle layer* of F_i . Two-step extensions are the case $L = 2$.

Lemma 9.1. *There is a smooth identification $E \cong (E_0, \bar{\partial}_0 + N)$, preserving the socle filtration, in which $N = \sum \beta_{ij}$ with $\beta_{ij} \in \Omega^{0,1}(\text{Hom}(F_j, F_i))$ harmonic, and $\beta_{ij} \neq 0$ only if the layer of F_i is smaller than that of F_j . The directed graph Γ with an edge $i \rightarrow j$ for each nonzero β_{ij} is connected.*

Proof. Refine the socle filtration to a Jordan–Hölder filtration and apply Lemma 6.1. Each layer occupies a consecutive range of Jordan–Hölder indices, and the corresponding diagonal block of N is the normal form of the polystable, hence split, subquotient S_k/S_{k-1} . Its entries between adjacent indices are harmonic representatives of zero classes, so they vanish. Suppose the entries of index gap less than σ inside a layer vanish. Then the subquotient on indices $a, \dots, a + \sigma$ is the direct sum of the intermediate factors and the extension of $F_{a+\sigma}$ by F_a given by the harmonic entry. Being a subquotient of a polystable bundle, it is split, so that entry vanishes as well. Connectedness is equivalent to simplicity by the argument of Theorem 3.1. $\square$

A factor in layer $k \geq 2$ has a nonzero entry from layer $k - 1$. Otherwise it would be a holomorphic subbundle of E/S_{k-2} and would lie in layer $k - 1$.

9.2. Diagonal metrics and their correction. For $g = \text{diag}(d_i \text{Id}_{F_i})$ with $d_i > 0$ and $\prod_i d_i^{r_i} = 1$, put $N_g = gNg^{-1}$, $h_g = g^*h_0$,

$$k_{ij}(g) = \|(N_g)_{ij}\|_{L^2}^2, \quad M_i p_i(g) = \sum_j k_{ij}(g) - \sum_l k_{li}(g), \quad \mathfrak{e}_g = \sum_{ij} k_{ij}(g),$$

and let $\nu_1(g) \leq \dots \leq \nu_{m-1}(g)$ be the positive graph eigenvalues of Section 8 for these conductances. We call $\mathfrak{e}_g = \|N_g\|_{L^2}^2$ the *total conductance*, and write $D_g = \bar{\partial}_{0,\text{End}} + [N_g, \cdot]$ for the Dolbeault operator of $(E_0, \bar{\partial}_0 + N_g)$ on endomorphisms. The *diagonal flow* is (58), that is $\dot{g}g^{-1} = -\text{diag}(p_i(g) \text{Id}_{F_i})$.

The mean curvature of $(\bar{\partial}_0 + N_g, h_0)$ is $\kappa \text{Id} + C_g$, where $C_g = \sqrt{-1} \Lambda_\omega (N_g - N_g^*) \wedge (N_g - N_g^*)$, as in (32). Taking block traces shows $PC_g = \text{diag}(p_i(g) \text{Id}_{F_i})$. Theorem 3.1 holds with independent edge amplitudes, since its proof uses only harmonicity and $\eta = \|[N_g, \cdot]\| \leq C\sqrt{\mathfrak{e}_g}$:

$$(65) \quad \lambda_j(h_g) = \nu_j(g)(1 + O(\mathfrak{e}_g)), \quad 1 \leq j \leq m - 1,$$

uniformly as $\mathfrak{e}_g \rightarrow 0$, with no condition on the ratios of the conductances.

Lemma 9.2. *Along the diagonal flow, $\dot{\mathbf{c}} = -2 \sum_i M_i p_i^2$ and*

$$(66) \quad \dot{\mathbf{c}} \leq -\frac{8\mathbf{e}^2}{(L-1)^2|M|}, \quad \int_T^\infty \mathbf{e}^{3/2} d\tau \leq \frac{(L-1)^2|M|}{4} \sqrt{\mathbf{e}(T)}.$$

The flow exists for all time and preserves $\prod_i d_i^{\tau_i} = 1$.

Proof. Each conductance satisfies $\dot{k}_{ij} = -2(p_i - p_j)k_{ij}$, and summing gives the formula for $\dot{\mathbf{c}}$. Let φ_i be minus the layer of F_i , shifted so that $\sum_i M_i \varphi_i = 0$. Every edge $i \rightarrow j$ has $\varphi_i - \varphi_j \geq 1$, so

$$\mathbf{c} \leq \sum_{i \rightarrow j} k_{ij}(\varphi_i - \varphi_j) = \sum_i M_i p_i \varphi_i \leq \left(\sum_i M_i p_i^2 \right)^{1/2} \left(\sum_i M_i \varphi_i^2 \right)^{1/2}.$$

The shifted values lie in an interval of length $L-1$, so $\sum_i M_i \varphi_i^2 \leq (L-1)^2|M|/4$. This proves the differential inequality, and integration gives the integral bound. The total conductance decreases, so the p_i are bounded on finite intervals, and the positive d_i cannot reach zero or infinity in finite time. Since $\sum_i r_i p_i = 0$, the weighted determinant is preserved. $\square$

Lemma 9.3. *For sufficiently small $\mathbf{e}_g$, set $v_g = RQC_g$, $k_g = \exp(-v_g/2)$, and let $\hat{h}_g = (k_g g)^* h_0$ be the corrected diagonal metric. Along the diagonal flow the residual (49) satisfies $\|\mathcal{E}_{\hat{h}_g}\|_{L^\infty, \text{op}} \leq A\mathbf{e}_g^{3/2}$. Moreover $e^{-C\mathbf{e}_g} h_g \leq \hat{h}_g \leq e^{C\mathbf{e}_g} h_g$ and $\lambda_j(\hat{h}_g) = \nu_j(g)(1 + O(\mathbf{e}_g))$. The constants depend only on the fixed forms and h_0 .*

Proof. The split Laplacian preserves adjoints, so v_g is self-adjoint, and $v_g = O_{C^k}(\mathbf{e}_g)$ for every k . The first variation of mean curvature at the split metric is $s \mapsto \Delta_0 s$. Taylor expansion therefore gives, in the gauge $k_g g$,

$$\hat{K}_g = C_g - \Delta_0 v_g + O_{C^k}(\mathbf{e}_g^{3/2}) = PC_g + O_{C^k}(\mathbf{e}_g^{3/2}).$$

Uniformity follows from the fixed finite collection of harmonic forms: $N_g = O_{C^k}(\sqrt{\mathbf{e}_g})$, and the mixed terms between N_g and v_g are $O_{C^k}(\mathbf{e}_g^{3/2})$.

The diagonal flow gives $\dot{N}_g = -[PC_g, N_g] = O_{C^k}(\mathbf{e}_g^{3/2})$, hence $\dot{v}_g = O_{C^k}(\mathbf{e}_g^2)$. Although v_g need not commute with PC_g ,

$$\partial_\tau(k_g g)(k_g g)^{-1} = \dot{k}_g k_g^{-1} - k_g PC_g k_g^{-1} = -PC_g + O_{C^k}(\mathbf{e}_g^2).$$

Its sum with its adjoint is the transported metric derivative, so the terms $\mp 2PC_g$ cancel in the residual (49). This proves the residual bound. The metric comparison follows from $v_g = O(\mathbf{e}_g)$, and the spectral statement from (48) and (65). $\square$

9.3. Matching the flow at late times.

Lemma 9.4. *Let h solve (46) on E . For every large T ,*

$$h(T) = A_T g_T^* \tilde{h}_T, \quad A_T > 0, \quad \tilde{h}_T \longrightarrow h_0 \text{ in } C^\infty, \quad \mathbf{e}_{g_T} \longrightarrow 0,$$

where g_T is diagonal with $\prod_i d_i^{\tau_i} = 1$, and $\tilde{h}_T$ is a metric on $(E_0, \bar{\partial}_0 + N_{g_T})$. We call the maps g_T the diagonal gauges of h .

Proof. In the smooth unitary flow gauges write $D_T \rightarrow \bar{\partial}_0$ on E_0 . Each (E_0, D_T) is isomorphic to E .

Hom dimensions. Let F_i lie in layer k . A nonzero map from F_i to a semistable bundle of slope μ is injective with saturated image, and that image lies in the socle. Hence $\dim \text{Hom}(F_i, E/S_{k-1}) = 1$. At the split limit every such Hom space also has dimension one.

Splitting. For F_i in layer one, the Hom-bundle Laplacians converge smoothly, so their one-dimensional kernel projections do too. This gives holomorphic maps $\iota_{i,T} : F_i \rightarrow (E_0, D_T)$ converging to the standard inclusions. For large T their sum is fiberwise injective, and its image is the socle of (E_0, D_T) . The h_0 -orthogonal complement carries a quotient holomorphic structure on the remaining factors that converges smoothly to $\bar{\partial}_0$ and is isomorphic to E/S_1 . Repeating this L times gives a smooth identification $\sigma_T \rightarrow \text{Id}$ transporting D_T to $\bar{\partial}_0 + \alpha_T$. Here α_T is strictly block upper-triangular in layers, $\alpha_T \rightarrow 0$ smoothly, and the diagonal blocks are exactly $\bar{\partial}_{F_i}$.

Harmonic form. Apply the triangular Hodge induction in the proof of Lemma 6.1, with the filtration by layer gap in place of the Jordan–Hölder index gap. This is legitimate because α_T has no entries inside a layer, and products of entries of layer gaps a and b have layer gap $a + b$. At each step the entries of the next layer gap are replaced by their harmonic parts, using the gauge $\bar{\partial}_0^* \mathbb{G}$ of the current entries, where $\mathbb{G}$ is the Green operator of the Dolbeault Laplacian. The new product terms are polynomials in entries that tend to zero in C^∞ . Hence the gauge tends to the identity, and the resulting harmonic normal form $\bar{\partial}_0 + N'_T$ has $N'_T \rightarrow 0$.

Uniqueness up to diagonal maps. Let $\Phi : (E_0, \bar{\partial}_0 + N) \rightarrow (E_0, \bar{\partial}_0 + N'_T)$ be a holomorphic isomorphism. It preserves the socle filtration, which is the standard layer flag on both sides. On N'_T this is because the splitting sends the flag to the socle and the Hodge gauge is unipotent in layers. Inside a layer the gauge equation reduces to $\bar{\partial}_0 \Phi_{ij} = 0$, so Φ is $\text{diag}(z_i)$ there. Write $\Phi = Z + U$ with U strictly upper-triangular in layers. The (i, j) entry of $\bar{\partial}_0 \Phi + N'_T \Phi - \Phi N = 0$ is

$$\bar{\partial}_0 U_{ij} + z_j N'_{T,ij} - z_i N_{ij} + \sum_k (N'_{T,ik} U_{kj} - U_{ik} N_{kj}) = 0,$$

summed over k in layers strictly between those of i and j . By induction on the layer gap, the sum vanishes. The exact form $\bar{\partial}_0 U_{ij}$ then equals a harmonic form, so both vanish, and U_{ij} is a holomorphic section of $\text{Hom}(F_j, F_i)$, hence zero. Therefore $\Phi = Z$ and $N'_T = Z N Z^{-1}$.

A parallel diagonal unitary map removes the phases of the z_i . The moduli $d_i = |z_i|$, normalized so that $\prod_i d_i^{r_i} = 1$, define g_T ; this does not change N_{g_T} . The metric $\tilde{h}_T$ tends to h_0 , because the splitting and the Hodge gauge tend to the identity and the unitary maps are parallel. Also $\mathfrak{e}_{g_T} \rightarrow 0$, because $N'_T \rightarrow 0$. The resulting identification with E differs from g_T by an automorphism of E , which is scalar since E is simple. Its squared modulus is A_T . $\square$

9.4. The transfer theorem.

Theorem 9.5. *Start the flow (46) on E from any smooth metric. For every sufficiently large T there are a solution $g^{[T]}$ of the diagonal flow on $[T, \infty)$ and a number $\delta_T \rightarrow 0$ such that*

$$(67) \quad e^{-\delta_T} \nu_j(g^{[T]}(\tau)) \leq \lambda_j(h(\tau)) \leq e^{\delta_T} \nu_j(g^{[T]}(\tau)), \quad \tau \geq T, \quad 1 \leq j < m.$$

Moreover $\liminf \lambda_m(h(\tau)) > 0$, and in smooth unitary flow gauges the full small spectral projection converges in every $L^2 \rightarrow C^k$ norm to the projection onto $\mathcal{K}$.

Proof. At time T apply Lemma 9.4. Start the diagonal flow at g_T , and compare h with $A_T \hat{h}_{g^{[T]}(\tau)}$. The initial discrepancy is $O(\epsilon_T + \mathfrak{e}_{g_T})$, where $e^{-\epsilon_T} h_0 \leq \tilde{h}_T \leq e^{\epsilon_T} h_0$. Lemmas 9.2 and 9.3 bound the total residual by $O(\sqrt{\mathfrak{e}_{g_T}})$. The barriers (50) then give

$$(68) \quad e^{-b_T} A_T \hat{h}_{g^{[T]}(\tau)} \leq h(\tau) \leq e^{b_T} A_T \hat{h}_{g^{[T]}(\tau)}, \quad b_T = \epsilon_T + C \mathfrak{e}_{g_T} + \frac{A(L-1)^2 |M|}{4} \sqrt{\mathfrak{e}_{g_T}}.$$

Combine this with (48) and Lemma 9.3 to obtain (67) with $\delta_T \leq 4b_T + C \mathfrak{e}_{g_T}$. The remaining assertions follow by resolvent convergence, as in Theorem 7.1. $\square$

Corollary 9.6. *Fix $j < m$ and a positive function f with $f(\tau + \sigma)/f(\tau) \rightarrow 1$ for every fixed σ . Suppose every solution g of the diagonal flow satisfies $\nu_j(g(\tau))/f(\tau) \rightarrow c_j(g) > 0$. Then every solution of (46) has $\lambda_j(h(\tau))/f(\tau) \rightarrow c_j(h) > 0$, with $|\log(c_j(h)/c_j(g^{[T]}))| \leq \delta_T$.*

Proof. At one fixed large T , the comparison (67) gives positive finite bounds on the lower and upper limits ℓ, u of λ_j/f . For each larger T , the comparison on $[T, \infty)$ gives $e^{-\delta_T} c_j(g^{[T]}) \leq \ell \leq u \leq e^{\delta_T} c_j(g^{[T]})$. Hence $u/\ell \leq e^{2\delta_T}$ for all large T , and $u = \ell$. The shift condition makes the choice of time origin of $g^{[T]}$ immaterial. $\square$

Lemma 9.7. *Let g_τ be the diagonal gauges of Lemma 9.4 at time τ . For every edge and every large T ,*

$$e^{-2b_T - o_\tau(1)} \leq \frac{k_{ij}(g_\tau)}{k_{ij}(g^{[T]}(\tau))} \leq e^{2b_T + o_\tau(1)}.$$

Proof. Evaluate (68) and the representation $h(\tau) = A_\tau g_\tau^* \tilde{h}_\tau$ on a nonzero vector of F_i . Since g acts on F_i by the scalar d_i , and $\tilde{h}_g \in e^{\pm C\epsilon_g} h_g$, this gives $d_i(g_\tau)^2/d_i(g^{[T]}(\tau))^2$ up to the factor A_T/A_τ and $e^{\pm(b_T + o_\tau(1))}$. Dividing the bounds for the two endpoints of an edge cancels A_T/A_τ . $\square$

Two choices of the diagonal gauges g_τ , with $\tilde{h}, \tilde{h}' \rightarrow h_0$, have conductance ratios tending to one, by the same argument. The limits of conductances along g_τ in Section 10 therefore do not depend on these choices.

10. THE COMPLETE LEADING SMALL SPECTRUM

We keep the setting of Section 9: a compact curve, and a simple bundle E with pairwise nonisomorphic Jordan–Hölder factors, written in the normal form of Lemma 9.1 with graph Γ and masses $M_i = Vr_i$. We use the weight grading w , the tight subgraph $\mathcal{T}$, the slacks ε_α , the exponents $a_\alpha = 1 + \varepsilon_\alpha$, and the a -quotient graphs of Section 8. These depend only on Γ and the ranks. List the small eigenvalues as $\lambda_1 \leq \dots \leq \lambda_{m-1}$. Assign exponents to indices so that larger exponents take smaller indices: the first $d_{a_{\max}}$ indices receive $a_{\max}$, and so on. Write a_j for the exponent of index j .

Theorem 10.1. *Assume (Γ, M) is nondegenerate. For every smooth initial metric on E :*

- (1) $\tau^{a_j} \lambda_j(h(\tau)) \rightarrow c_j(h) > 0$ for every $j < m$.
- (2) *The coefficients with $a_j = 1$ are the positive eigenvalues of the tight subgraph with conductances u^* and masses M_i , with multiplicity. They do not depend on the initial metric. The value $\frac{1}{2}$ occurs with multiplicity at least the number of tight components with at least two vertices.*
- (3) *For every edge the limit $k_\alpha^\infty(h) = \lim \tau^{a_\alpha} k_\alpha(g_\tau)$ exists along the diagonal gauges of Lemma 9.4. For $a_j = a > 1$, the coefficients $c_j(h)$ are the positive eigenvalues of the a -quotient graph with conductances $k^\infty(h)$, with multiplicity. For every $a > 1$ with $d_a \geq 1$, their sum, and hence the largest of them, is unbounded as $h(0)$ ranges over metrics on the single bundle E .*
- (4) *In smooth unitary flow gauges, for each exponent a and each distinct value ρ among the coefficients with $a_j = a$, the spectral projection onto $\{\lambda_j : a_j = a, c_j(h) = \rho\}$ converges in every $L^2 \rightarrow C^k$ norm to the ρ -eigenspace of the a -quotient graph. This eigenspace is embedded in $\mathcal{K}$ as block-scalar endomorphisms constant on the contracted vertices. For $\rho = \frac{1}{2}$ at $a = 1$ it contains $\sum_{i \in \mathcal{C}} w_i \text{Id}_{F_i}$ for each tight component $\mathcal{C}$ with at least two vertices.*

Proof. Eigenvalues. By Theorems 8.3 and 8.5, every solution g of the diagonal flow has $\tau^{a_j} \nu_j(g(\tau)) \rightarrow \rho_j(g) > 0$. The exponent assigned to each index is the same for all solutions, and $f = \tau^{-a}$ satisfies the shift condition. Corollary 9.6 gives (1), with $|\log(c_j(h)/\rho_j(g^{[T]}))| \leq \delta_T$. At $a = 1$ the value $\rho_j(g)$ is the same for all g , which proves (2).

Edge limits. By (64), $\tau^{a_\alpha} k_\alpha(g^{[T]}(\tau)) \rightarrow k_\alpha^\infty(g^{[T]})$. Lemma 9.7 puts the lower and upper limits of $\tau^{a_\alpha} k_\alpha(g_\tau)$ within $e^{\pm 2b_T}$ of $k_\alpha^\infty(g^{[T]})$ for every large T . Since $b_T \rightarrow 0$, the limit $k_\alpha^\infty(h)$ exists and equals $\lim_T k_\alpha^\infty(g^{[T]})$. The positive eigenvalues of each a -quotient graph depend continuously on the conductances, so $c_j(h)$ is the corresponding eigenvalue for $k^\infty(h)$.

Unboundedness. Fix $a > 1$ with $d_a \geq 1$ and constants θ_C with $\sum_C M_C \theta_C = 0$. For a large s_0 , let g_* be the diagonal map with logarithms $ws_0 + \zeta^* + \sum_C \theta_C \mathbf{1}_C$, and set $h(0) = g_*^* h_0$. This is a metric on E .

- The diagonal flow from g_* at time zero is a time shift of the solution considered at the end of the proof of Theorem 8.5. Hence its constants satisfy $Z_C = \theta_C + O(e^{4 \max |\theta_C| - \varepsilon_* s_0})$.
- Its total conductance is $\mathfrak{e}_0 \leq C e^{-s_0} + C e^{4 \max |\theta_C|} e^{-(1+\varepsilon_*) s_0}$.
- Compare h on $[0, \infty)$ with the corrected diagonal metrics along the diagonal flow from g_* . The initial discrepancy is $O(\mathfrak{e}_0)$, and Lemma 9.2 bounds the total residual by $O(\sqrt{\mathfrak{e}_0})$. The barriers give $e^{-b} \widehat{h}_{g(\tau)} \leq h(\tau) \leq e^b \widehat{h}_{g(\tau)}$ for all $\tau \geq 0$, with $b = O(\mathfrak{e}_0 + \sqrt{\mathfrak{e}_0})$ and constants independent of g_* .

Therefore $c_j(h) \in e^{\pm 4b} \rho_j(g)$ for every j . Take $s_0 \gg 4 \max |\theta_C| / \varepsilon_*$. Then Theorem 8.5(3) shows that the sum of the coefficients at exponent a takes arbitrarily large values as (θ_C) varies.

Spectral subspaces. Let u be a normalized eigensection with $a_j = a$ and $c_j(h) = \rho$, and transport it to $(\tilde{\partial}_0 + N_{g_\tau}, \tilde{h}_\tau)$. Write $u = x + y$ with $x \in \mathcal{K}$ and $y \perp \mathcal{K}$ in h_0 .

- Changing $\tilde{h}_\tau$ to h_0 alters pointwise norms by a factor $1 + o(1)$. So $\|D_{g_\tau} u\|_{h_0}^2 \leq \lambda(1 + o(1))$ with $\lambda = O(\tau^{-a})$. By (25), which uses the cancellation only in h_0 ,

$$\|[N_{g_\tau}, x]\|^2 + \|y\|^2 \leq C\lambda.$$

- For edges with $a_\alpha < a$, $k_\alpha(g_\tau) \geq c\tau^{-a_\alpha}$ by $k_\alpha^\infty(h) > 0$. So every limit x_∞ of x is constant on the contracted vertices. The identifications with the diagonal gauges consist of a map tending to the identity followed by a parallel diagonal unitary map, which fixes $\mathcal{K}$ pointwise. Full-projection convergence (Theorem 9.5) therefore gives $y \rightarrow 0$ in every C^k norm.
- Test the weak eigenvalue equation against a fixed block-scalar z constant on the contracted vertices. Then $\|[N_{g_\tau}, z]\|^2 = O(\tau^{-a})$. Replacing $\tilde{h}_\tau$ by h_0 costs $o(\tau^{-a})$. In h_0 the complementary term is $\langle [N_{g_\tau}, y], [N_{g_\tau}, z] \rangle$, since $D_0^*[N, z] = 0$. It is at most

$$\|N_{g_\tau}\|_{L^\infty} \|y\| \|[N_{g_\tau}, z]\| = O(\tau^{-1/2}) O(\tau^{-a/2}) O(\tau^{-a/2}) = o(\tau^{-a}),$$

using $\mathfrak{e}_{g_\tau} = O(\tau^{-1})$, which follows from the decay bound of Lemma 9.2 together with Lemma 9.7.

Multiply by τ^a and pass to the limit, using the edge limits. Edges with $a_\alpha < a$ do not contribute, since z is constant across them, and edges with $a_\alpha > a$ contribute $o(1)$. So x_∞ is a ρ -eigenvector of the a -quotient graph. The number of eigenvalues in this group equals the multiplicity of ρ . Limits of orthonormal eigensections remain orthonormal because $y \rightarrow 0$. Compactness in $\mathcal{K}$ and smooth convergence of y give the stated $L^2 \rightarrow C^k$ convergence. The last assertion of (4) is Theorem 8.5(2). $\square$

Corollary 10.2. *Under the hypotheses of Theorem 10.1, write $h(\tau) = A_\tau g_\tau^* \tilde{h}_\tau$ with the diagonal gauges g_τ of Lemma 9.4 and $\tilde{h}_\tau \rightarrow h_0$. The limits $Z_C(h) = \lim_T Z_C(g^{[T]})$ exist, and, up to a common scalar,*

$$\log g_\tau = w \log \tau + y_\infty(h) + o(1), \quad y_\infty(h) = \zeta^* + \sum_C Z_C(h) \mathbf{1}_C.$$

Proof. The edge limits give convergence of $\log d_i - \log d_j - (w_i - w_j) \log \tau$ along every edge. Since the graph is connected, all such differences converge. $\square$

Haiden, Katzarkov, Kontsevich, and Pandit [HKKP18, Theorem 1.1] and Lam [Lam26, Theorem 1.1] describe the metric growth up to a bounded term. Corollary 10.2 shows that in the nondegenerate case this bounded term converges. The limits $Z_C(h)$ exist because the edge limits converge and the quotient of Γ by the tight components is connected. So the limit depends on the initial metric only through the constants $Z_C(h)$.

11. EXAMPLES ALONG THE FLOW

We work on a compact curve X of genus at least two. There, stable bundles of every rank and degree exist. For stable F_i, F_j of equal slope that are not isomorphic, $h^0(\text{Hom}(F_j, F_i)) = 0$ and Riemann–Roch gives $h^1(\text{Hom}(F_j, F_i)) = r_i r_j (g(X) - 1) > 0$. Every graph Γ of the kind considered below is therefore realized by nonzero harmonic entries.

11.1. All edges tight. When every edge is tight and $(\Gamma, (r_i))$ is nondegenerate, there is one tight component and every small eigenvalue decays like τ^{-1} . For two-step extensions this is the *positive balance condition*. It asks for positive numbers σ_{ij} on the edges with prescribed margins,

$$(69) \quad \sum_j \sigma_{ij} = \frac{M_i M_J}{2|M|} \quad (i \in I), \quad \sum_i \sigma_{ij} = \frac{M_j M_I}{2|M|} \quad (j \in J),$$

where $M_I = \sum_{i \in I} M_i$ and $M_J = \sum_{j \in J} M_j$. Feasibility is a classical matrix-scaling question [SK67, RS89].

Condition (69) holds exactly when every edge is tight and $(\Gamma, (r_i))$ is nondegenerate. Let $\bar{w}$ take the value $-M_J/(2|M|)$ on I and $M_I/(2|M|)$ on J . It is admissible, with equality on every edge, and (69) is the multiplier equation (59) for $\bar{w}$ with multipliers σ . A pair satisfying these optimality conditions of a convex program characterizes its minimizer, so $\bar{w}$ is the weight grading, every edge is tight, and $(\Gamma, (r_i))$ is nondegenerate. Conversely, if every edge is tight and Γ is connected and bipartite, the differences $w_j - w_i = \frac{1}{2}$ and $\sum_i M_i w_i = 0$ force $w = \bar{w}$, and positive multipliers satisfy (69).

Corollary 11.1. *Assume (69). There are unique conductances $k_{ij}^* = \|\beta_{ij}\|^2 e^{2(y_i^* - y_j^*)}$ with these margins, normalized by $\sum_i M_i y_i^* = 0$. Then $\tau \lambda_j(h(\tau)) \rightarrow \rho_j$ for every initial metric, where $\rho_1 \leq \dots \leq \rho_{m-1}$ are the positive eigenvalues of the graph with conductances k^* and masses M_i . The spectral projections onto the eigenvalues with a common limit converge to the corresponding graph eigenspaces.*

Proof. This is Theorem 10.1 with $\mathcal{T} = \Gamma_1$ and $u^* = k^*$. $\square$

Example 11.2. Take four distinct degree-zero line bundles and all four entries $\beta_{12}, \beta_{14}, \beta_{32}, \beta_{34}$ nonzero, with $V = 1$. The quantity

$$\Theta = \frac{\|\beta_{12}\|^2 \|\beta_{34}\|^2}{\|\beta_{14}\|^2 \|\beta_{32}\|^2}$$

is invariant under diagonal changes of the factor metrics. The balanced matrix, with rows 1, 3 and columns 2, 4, is

$$\begin{pmatrix} x & y \\ y & x \end{pmatrix}, \quad x = \frac{\sqrt{\Theta}}{4(1 + \sqrt{\Theta})}, \quad y = \frac{1}{4(1 + \sqrt{\Theta})}.$$

Its positive graph eigenvalues are $2x$, $2y$ and $\frac{1}{2}$. Thus every initial metric gives

$$\lim_{\tau \rightarrow \infty} \tau(\lambda_1, \lambda_2, \lambda_3) = \left(\frac{\min(1, \sqrt{\Theta})}{2(1 + \sqrt{\Theta})}, \frac{\max(1, \sqrt{\Theta})}{2(1 + \sqrt{\Theta})}, \frac{1}{2} \right).$$

The constants retain the invariant Θ of the extension norms, but not the extension classes themselves. The tight subgraph is a four-cycle, so u^* depends on the norms. The value $\frac{1}{2}$ has eigenvector $w = \frac{1}{4}(-1, 1, -1, 1)$.

The three-factor extension $0 \rightarrow L_1 \oplus L_3 \rightarrow E \rightarrow L_2 \rightarrow 0$ with $\beta_{12}, \beta_{32} \neq 0$ and $V = 1$ is of this kind. Its weight grading is $(-\frac{1}{6}, \frac{1}{3}, -\frac{1}{6})$, and both multipliers equal $\frac{1}{6}$. So

$$\lambda_1 \sim \frac{1}{6\tau}, \quad \lambda_2 \sim \frac{1}{2\tau},$$

with limiting lines $\mathbb{C} \text{diag}(1, 0, -1)$ and $\mathbb{C} \text{diag}(1, -2, 1)$.

11.2. A leading coefficient depending on the initial metric. Let $V = 1$, and choose pairwise nonisomorphic stable degree-zero bundles F_1, F_2, F_3, F_4 of ranks $(2, 1, 1, 2)$, with nonzero harmonic entries $\beta_{12}, \beta_{32}, \beta_{34}$ and no others.

The weight grading is $w = (-\frac{1}{6}, \frac{1}{3}, -\frac{1}{3}, \frac{1}{6})$. The edges 12 and 34 are tight with multipliers $\frac{1}{3}$. The edge 32 has slack $\frac{1}{3}$, hence exponent $\frac{4}{3}$; the edge 32 is slack because $M_1 M_4 = 4 > 1 = M_2 M_3$. The multipliers on the tight edges are positive, so $(\Gamma, (r_i))$ is nondegenerate. This is the region treated in [HKP23, (3.24)–(3.26)].

Theorem 10.1 gives:

- two eigenvalues asymptotic to $1/(2\tau)$ (the tight components $\{1, 2\}$ and $\{3, 4\}$, each contributing $\frac{1}{2}$), with limiting subspace $\text{span}\{\text{diag}(1, -2, 0, 0), \text{diag}(0, 0, 2, -1)\}$;
- one eigenvalue $\lambda_1 \sim c(h(0))\tau^{-4/3}$, with limiting line $\mathbb{C} \text{diag}(1, 1, -1, -1)$ and $c(h(0)) = \frac{2}{3}k_{32}^\infty(h)$.

The coefficient is unbounded on the metrics of one bundle. The explicit family below makes this quantitative.

Proposition 11.3. *Normalize the three entries to have norm one, and for $\rho > 0$ put*

$$g_\rho = \text{diag}(\rho^{1/2} \text{Id}_{F_1}, \rho^{-1/2} \text{Id}_{F_2}, \rho^{1/2} \text{Id}_{F_3}, \rho^{-1/2} \text{Id}_{F_4}), \quad h_\rho = g_\rho^* h_0.$$

Then the flows starting at h_ρ satisfy

$$c(h_\rho) = \frac{2}{27} \rho^{-2/3} (1 + O(\rho)), \quad \rho \downarrow 0.$$

Proof. Write $k_1 = k_{12}$, $k_2 = k_{32}$, $k_3 = k_{34}$. The masses give $p = (k_1/2, -k_1 - k_2, k_2 + k_3, -k_3/2)$ and

$$\dot{k}_1 = -3k_1^2 - 2k_1 k_2, \quad \dot{k}_2 = -2(k_1 + k_3)k_2 - 4k_2^2, \quad \dot{k}_3 = -3k_3^2 - 2k_2 k_3.$$

The symmetric solution $k_1 = k_3 = u$, with initial values u_0 and $r_0 = k_2(T)/u_0$, is explicit. In the time s with $ds/d\tau = u$ and $s(T) = 0$, put $D(s) = 1 + 2r_0 - 2r_0 e^{-s}$. Then

$$r(s) = \frac{r_0 e^{-s}}{D(s)}, \quad u(s) = \frac{u_0 e^{-3s}}{D(s)}, \quad \tau - T = \frac{1}{u_0} \left(\frac{(1 + 2r_0)(e^{3s} - 1)}{3} - r_0(e^{2s} - 1) \right),$$

so

$$\tau^{4/3}k_2 \longrightarrow B = \frac{r_0}{3^{4/3}u_0^{1/3}(1+2r_0)^{2/3}}.$$

For h_ρ all three initial conductances are ρ^2 , so $u_0 = \rho^2$ and $r_0 = 1$, and the diagonal flow gives the coefficient $\frac{2}{3}B = 2/(27\rho^{2/3})$. As in the proof of Theorem 10.1, compare the flow from h_ρ with the corrected diagonal metrics along this solution, from time zero. Its initial discrepancy from h_ρ is $O(\rho^2)$, and its total residual is $O(\rho)$. The barriers put the actual coefficient within $e^{\pm C\rho}$ of the diagonal one. $\square$

11.3. A degenerate example: logarithmic separation. Let $L_1, \dots, L_4$ be pairwise non-isomorphic degree-zero line bundles, with $V = 1$, and take the zigzag $0 \rightarrow L_1 \oplus L_3 \rightarrow E \rightarrow L_2 \oplus L_4 \rightarrow 0$ with exactly $\beta_{12}, \beta_{32}, \beta_{34}$ nonzero. The masses are equal, so $M_1M_4 = M_2M_3$ and $(\Gamma, (r_i))$ is degenerate. The middle edge is tight, but its multiplier is forced to vanish. Theorem 10.1 does not apply. The diagonal flow is the scalar zigzag of [HKKP23, Example 5.13], and the transfer theorem still applies.

Theorem 11.4. *For every smooth initial metric,*

$$\lambda_1(h(\tau)) \sim \frac{1}{2\tau \log \tau}, \quad \lambda_2(h(\tau)) \sim \lambda_3(h(\tau)) \sim \frac{1}{2\tau}, \quad \liminf_{\tau \rightarrow \infty} \lambda_4(h(\tau)) > 0.$$

The first line tends to $\mathbb{C} \operatorname{diag}(1, 1, -1, -1)$, and the second and third together tend to $\operatorname{span}_{\mathbb{C}}\{\operatorname{diag}(1, -1, 0, 0), \operatorname{diag}(0, 0, 1, -1)\}$.

The constants do not depend on the extension forms or the initial metric.

Proof. With $k_1 = k_{12}$, $k_2 = k_{32}$, $k_3 = k_{34}$, the diagonal flow is

$$\dot{k}_1 = -2k_1(2k_1 + k_2), \quad \dot{k}_2 = -2k_2(k_1 + 2k_2 + k_3), \quad \dot{k}_3 = -2k_3(2k_3 + k_2).$$

Put $u = \sqrt{k_1 k_3}$, $r = k_2/u$, $z = \sqrt{k_1/k_3}$, $\xi = (z-1)/(z+1)$, and $ds/d\tau = 2u$ with $s(T) = 0$. Then

$$\frac{dr}{ds} = -r^2, \quad \frac{d\xi}{ds} = -2\xi, \quad \frac{d \log u}{ds} = -\left(2 + \frac{4\xi^2}{1-\xi^2} + r\right).$$

Hence $r = (s + r_0^{-1})^{-1}$ and $\xi = \xi_0 e^{-2s}$ with $|\xi_0| < 1$. There is $C > 0$ with $u = C e^{-2s}(s + r_0^{-1})^{-1}(1 + O(e^{-4s}))$, and $\tau \sim e^{2s}(s + r_0^{-1})/(4C)$. Therefore

$$\tau k_1 \rightarrow \frac{1}{4}, \quad \tau k_3 \rightarrow \frac{1}{4}, \quad \tau \log \tau k_2 \rightarrow \frac{1}{2},$$

for every positive initial condition. The pairs $\{1, 2\}$ and $\{3, 4\}$ give two graph eigenvalues asymptotic to $1/(2\tau)$. On vectors constant on the pairs, the remaining eigenvalue is asymptotic to k_2 (quotient masses 2, 2). Eliminating the internal directions changes it by $O(k_2^2/\min(k_1, k_3)) = o(k_2)$. So $\nu_1 \sim (2\tau \log \tau)^{-1}$ and $\nu_2 \sim \nu_3 \sim (2\tau)^{-1}$.

Corollary 9.6 applies with $f = (\tau \log \tau)^{-1}$ and $f = \tau^{-1}$, which proves the eigenvalue statements. For the first line, a normalized eigensection has Rayleigh quotient $O((\tau \log \tau)^{-1})$. Along the diagonal gauges g_τ , Lemma 9.7 gives $k_1, k_3 \sim 1/(4\tau)$. With the form bound (25), this forces $x_1 = x_2$ and $x_3 = x_4$ in every limit. The complementary two-dimensional space follows by orthogonality. Full-projection convergence upgrades these identifications to $L^2 \rightarrow C^k$ convergence. $\square$

The four-factor first eigensection separates $L_1 \oplus L_2$ from $L_3 \oplus L_4$, and the faster subspace separates the members of each pair. Adjoining the identity to the full limiting small subspace recovers all four summand projections.

12. TWO-STEP EXTENSIONS IN HIGHER DIMENSIONS

Let (X, ω) be a compact connected Kähler manifold of complex dimension $n \geq 1$, with the conventions of Section 1. Let $F_1, \dots, F_m$ be pairwise nonisomorphic stable vector bundles of common slope, with Hermitian–Einstein metrics h_i . Partition the indices as $I \sqcup J$, and let

$$0 \longrightarrow S = \bigoplus_{i \in I} F_i \longrightarrow E \longrightarrow U = \bigoplus_{j \in J} F_j \longrightarrow 0$$

be an extension, with harmonic representative $N = \sum \beta_{ij}$. Here each entry β_{ij} is a harmonic form in $\Omega^{0,1}(\text{Hom}(F_j, F_i))$. Assume that the graph Γ of nonzero entries is connected.

Theorem 12.1. *Let E be as above, with masses $M_i = V \text{rk } F_i$ and conductances $k_{ij} = \|(gNg^{-1})_{ij}\|_{L^2}^2$. Along the normalized flow from any smooth initial metric on E :*

- (1) *every small eigenvalue satisfies the two-sided comparison (67) with the diagonal flow, with $\delta_T \rightarrow 0$, and the full small spectral projection converges to $\mathcal{K}$ in every $L^2 \rightarrow C^k$ norm;*
- (2) *if $(\Gamma, (r_i))$ is nondegenerate, conclusions (1)–(4) of Theorem 10.1 hold;*
- (3) *for two factors, $\tau\lambda_1(h(\tau)) \rightarrow \frac{1}{2}$, as in Theorem 7.1.*

Corollaries 9.6 and 11.1 and the statements of Section 11 hold on X whenever the required bundles and nonzero harmonic entries exist there.

The fixed-metric results of Sections 2 and 3, including Theorem 4.1, also hold on X for two-step extensions. Section 2 is already stated there. For more than two socle layers the families need not be integrable when $n \geq 2$. On $\Sigma_1 \times \Sigma_2$, for instance, entries β_{12} and β_{23} pulled back from the two factors have a nonzero product class in $H^{0,2}(\text{Hom}(F_3, F_1))$.

Proof. We check each input used on curves.

Integrability. Harmonic forms are $\bar{\partial}_0$ -closed, and $\beta_{ij} \wedge \beta_{kl}$ requires $j = k$, which cannot occur since $j \in J$ and $k \in I$. So $\bar{\partial}_0 N + N \wedge N = 0$, and the same holds for gNg^{-1} and every entrywise rescaling.

Mean curvature. The Chern connection of $(\bar{\partial}_0 + N, h_0)$ is $A_0 + N - N^*$. Since $\bar{\partial}_0^* N = 0$, the Kähler identity $[\Lambda, \partial_0] = \sqrt{-1} \bar{\partial}_0^*$ gives $\Lambda \partial_0 N = 0$, and likewise for $\bar{\partial}_0 N^*$. Hence $K - \kappa \text{Id} = \sqrt{-1} \Lambda(N - N^*) \wedge (N - N^*)$. In a local unitary coframe with $N = \sum_\nu n_\nu \bar{\theta}^\nu$, this is $\sum_\nu [n_\nu, n_\nu^*]$, as in (9). Its block traces give the same p_i as on a curve.

Fixed-metric comparison. The proofs of Theorem 3.1 and Proposition 3.3 use the cancellation $D_0^*[N, x] = [\bar{\partial}_0^* N, x] = 0$, the spectral gap of Δ_0 on QH , elliptic estimates, and Sobolev embedding. All of these hold on X .

Order comparison. Propositions 4.3 and 4.5 of [HKKP18], used in (50), are stated and proved on compact Kähler manifolds of any dimension, by a pointwise maximum principle.

Correction. The first variation of mean curvature at a Hermitian–Einstein metric is $\Delta_\partial = \Delta_{\bar{\partial}}$ on endomorphisms: the difference is $[K, \cdot]$, which vanishes there. The mixed terms in Lemma 9.3 are $O(\epsilon^{3/2})$ uniformly. So that lemma and Lemma 9.2 hold unchanged.

Convergence of the flow. Every Jordan–Hölder filtration of E has graded object isomorphic to $S \oplus U$. A saturated stable subsheaf of slope μ in E is reflexive, hence isomorphic to some F_k , and $\text{Hom}(F_k, E) = 0$ for $k \in J$ since each extension class is nonzero. So the Harder–Narasimhan–Seshadri graded object is locally free, and its singular set is empty. By Sibley–Wentworth [SW15, Theorem 1.1], the flow converges modulo unitary gauge, smoothly on all of X , to an admissible Yang–Mills connection A_∞ on a bundle isomorphic to $S \oplus U$. They credit the smooth convergence to Hong and Tian [HT04], and the identification of the limit to Jacob and Sibley [Jac15, Sib15].

A_∞ is smooth and integrable. Its Hermitian–Einstein summands have constants given by their slopes, and every holomorphic summand of $\bigoplus F_i$ has slope μ . So A_∞ is Hermitian–Einstein, and it is gauge equivalent to the product connection of h_0 . For each k , the infimum over unitary gauges of the C^k distance to A_∞ tends to zero. Choosing near-optimal gauges with k tending slowly to infinity gives, for every large τ , holomorphic isometries from $(E, h(\tau))$ to $(E_0, h_0, \bar{\partial}^{(\tau)})$ with $\bar{\partial}^{(\tau)} \rightarrow \bar{\partial}_0$ in C^∞ . This is the input used on curves.

Matching. In Lemmas 7.3 and 9.4, the transported extension entry α_T is $\bar{\partial}$ -closed: in the block computation of $D_T^2 = 0$, the product $\alpha_T \wedge \alpha_T$ lands in $\text{Hom}(U, S) \circ \text{Hom}(U, S) = 0$. Hodge theory on X gives $\alpha_T = H\alpha_T + \bar{\partial}v_T$ with $v_T \rightarrow 0$ in C^∞ . Uniqueness up to diagonal maps uses only L^2 -orthogonality of harmonic and exact forms and $H^0(\text{Hom}(U, S)) = 0$.

The remaining arguments are finite-dimensional or use only these inputs. $\square$

Example 12.2. Let $X = \Sigma \times Y$, where Σ is a curve of genus at least two and Y is a connected compact Kähler manifold, with a product Kähler form.

- For stable F on Σ , the pullback of a Hermitian–Einstein metric is Hermitian–Einstein, since Λ_ω of a form pulled back from Σ is Λ_{ω_Σ} of that form. By the Künneth formula $H^0(X, \text{End } \pi^*F) = \mathbb{C}$, so π^*F is stable.
- Nonisomorphic stable bundles of equal slope pull back to such bundles, and

$$H^1(X, \text{Hom}(\pi^*G, \pi^*F)) \supset H^1(\Sigma, \text{Hom}(G, F)) \neq 0.$$

- Harmonic forms pull back to harmonic forms.

Every example of Section 11 therefore occurs on X , with arbitrary initial metrics on X . The same holds for the sharpness families of Section 15. For the explicit constants, $V = 1$ means $\text{Vol}(\Sigma) \text{Vol}(Y) = 1$.

For extensions with more than two steps in dimension $n \geq 2$, the Maurer–Cartan equation prevents all entries from being harmonic. A natural replacement is the gauge $\bar{\partial}_0^* N = 0$. In it, the entry N_{13} acquires the cup-product correction $-\bar{\partial}_0^* \mathbb{G}(N_{12} \wedge N_{23})$, with $\mathbb{G}$ the Green operator, which exists only when $[N_{12}] \cup [N_{23}] = 0$. We do not treat this case.

13. REPEATED STABLE SUMMANDS

The relative estimate also applies when some stable summands are isomorphic. The limiting kernel then contains matrices acting between the repeated copies, and the comparison operator is a commutator quadratic form on this larger space.

Retain the curve, scaling exponents, and harmonic upper-triangular family (17), but allow repetitions among the stable equal-slope bundles $F_1, \dots, F_m$. Choose pairwise nonisomorphic representatives $H_1, \dots, H_s$ and identifications

$$(70) \quad G = \bigoplus_{i=1}^m F_i \simeq \bigoplus_{\alpha=1}^s H_\alpha \otimes \mathbb{C}^{m_\alpha}, \quad \sum_{\alpha} m_\alpha = m.$$

Here $H_\alpha \otimes \mathbb{C}^{m_\alpha}$ is the direct sum of all copies of H_α , called its isotypical component. Fix a Hermitian–Einstein metric on each H_α and give its copies the transported metric. Thus the product metric on each isotypical component is $h_{H_\alpha} \otimes I$. Write $r_\alpha = \text{rk } H_\alpha$.

Stability at equal slope gives

$$(71) \quad \mathcal{A} := H^0(X, \text{End } G) = \bigoplus_{\alpha=1}^s \text{Id}_{H_\alpha} \otimes M_{m_\alpha}(\mathbb{C}).$$

Every element of $\mathcal{A}$ is parallel for the split Chern connection: under the chosen identifications its nonzero entries are constant multiples of the identity of a stable representative. Consequently

$$(72) \quad \begin{aligned} \mathcal{K} &:= H^0(X, \text{End}_0 G) = \left\{ (X_\alpha)_\alpha : \sum_\alpha r_\alpha \text{tr } X_\alpha = 0 \right\}, \\ d &:= \dim_{\mathbb{C}} \mathcal{K} = \sum_\alpha m_\alpha^2 - 1, \quad \|X\|^2 = V \sum_\alpha r_\alpha \text{tr}(X_\alpha^* X_\alpha). \end{aligned}$$

Use $D_t = D_0 + B_t$, $B_t x = [N_t, x]$, $\Delta_t = D_t^* D_t$, P, Q, γ, η_t , and p_* as before, now with this kernel. Define the nonnegative self-adjoint operator L_t on $\mathcal{K}$ by

$$(73) \quad \langle x, L_t x \rangle = A_t(x) := \|[N_t, x]\|_{L^2}^2, \quad L_t = P B_t^* B_t P|_{\mathcal{K}}.$$

Let $\nu_1(t) \leq \dots \leq \nu_d(t)$ denote its eigenvalues, including zeros. Unlike (20), this form may have cross terms between extension entries contributing to the same matrix block.

Theorem 13.1 (Comparison with the commutator form). *If $\eta_t \leq \sqrt{\gamma}/4$ and $\gamma_t = (\sqrt{\gamma} - \eta_t)^2$, then exactly d eigenvalues of Δ_t lie below $\gamma/4$, counting zeros, and*

$$(74) \quad \left(1 - \frac{2\eta_t^2}{\gamma_t}\right) \nu_j(t) \leq \lambda_j(t) \leq \nu_j(t), \quad 1 \leq j \leq d, \quad \lambda_{d+1}(t) \geq \frac{\gamma}{4}.$$

In particular, $\lambda_j(t)/\nu_j(t) = 1 + O(t^{2p_*})$ for every positive $\nu_j(t)$, uniformly over these d modes. For every $t > 0$, the trace-free holomorphic endomorphism space of E_t has dimension

$$(75) \quad \dim_{\mathbb{C}} \{x \in \mathcal{K} : [N_t, x] = 0\}.$$

Thus E_t is simple if and only if the only elements of $\mathcal{A}$ commuting with N_t are scalar identities.

For a unit small eigensection with eigenvalue λ , and for the orthogonal projection Π_t onto the full d -dimensional small spectral subspace, one has, for every $k \geq 0$,

$$(76) \quad \|Qu\|_{C^k} \leq C_k t^{p_*} \sqrt{\lambda}, \quad \|\Pi_t - P\|_{L^2 \rightarrow C^k} = O(t^{2p_*}).$$

Proof. Multiplication on either side by a parallel endomorphism commutes with the split Dolbeault Laplacian on bundle-valued forms. Since N_t is harmonic and $x \in \mathcal{K}$ is parallel, $B_t x = [N_t, x]$ is harmonic. In particular,

$$D_0^* B_t x = 0 \quad (x \in \mathcal{K}).$$

For $u = x + y$, $x \in \mathcal{K}$ and $y \in QH^1$, the same gap estimate and cross-term bound as in (24) give

$$\|D_t u\|^2 \geq \left(1 - \frac{2\eta_t^2}{\gamma_t}\right) A_t(x) + \frac{\gamma_t}{2} \|y\|^2.$$

Also $\nu_d(t) \leq \eta_t^2 \leq \gamma/16$. The min-max argument in Theorem 3.1 therefore applies with d in place of $m - 1$ and proves (74). No commutativity of $\mathcal{A}$ is used in this argument.

The displayed lower bound identifies the kernel for small t with $\{x \in \mathcal{K} : B_t x = 0\}$. The parallel diagonal map g_t preserves $\mathcal{A}$ by conjugation, preserves trace, and satisfies $N_t = g_t N_1 g_t^{-1}$. It therefore identifies the commutants for different positive parameters. Since the bundles E_t are also holomorphically isomorphic, the kernel dimension formula holds for all $t > 0$.

To check smooth convergence, the complementary equation is still (27), with $R_t = QB_t^*B_tP$ and $\|R_tx\| \leq \eta_t\sqrt{A_t(x)}$. Every B_tx lies in the fixed finite-dimensional space of harmonic $(0,1)$ -forms with values in $\text{End}_0 G$. Equivalence of norms on that space gives

$$\|B_tx\|_{H^\ell} \leq C_\ell\sqrt{A_t(x)}, \quad \|R_tx\|_{H^\ell} \leq C_\ell t^{p*}\sqrt{A_t(x)}.$$

This estimate remains valid in the presence of cancellations between matrix entries. The uniform complementary inverse estimates and Sobolev embedding used in Proposition 3.3 now give the first assertion of (76). The largest small eigenvalue is $O(t^{2p*})$; applying that assertion to an orthonormal basis of the d small eigensections and using the projection formula for the resulting graph over $\mathcal{K}$ proves the second assertion. $\square$

When all $m_\alpha = 1$, (73) is exactly the weighted graph form. For repeated factors, connectedness of that graph is insufficient for simplicity. For example, take $G = H \oplus H$ with H stable and $N_t = t\beta E_{12}$, where β is a nonzero harmonic scalar $(0,1)$ -form. The parallel nilpotent endomorphism E_{12} commutes with N_t , so this connected two-vertex family is not simple. Here E_{ij} denotes the identity map from the j th copy of H to the i th, extended by zero on the other blocks.

The mean-curvature calculation (32) uses only harmonicity and the common Einstein constant, so it also applies with repeated factors. In particular, the transported metrics $g_t^*h_0$ have uniform mean-curvature error $O(t^{2p*})$ on the fixed bundle E_1 .

13.1. Recovery of factors and multiplicities. After adjoining the scalar identity, the limiting small subspace is the algebra $\mathcal{A}$ in (71). Its minimal nonzero central idempotents are the projections onto the isotypical components. The corresponding ideals have dimensions m_α^2 , so their matrix sizes determine the multiplicities. Within such an ideal, a minimal nonzero idempotent has rank one on the multiplicity space and its holomorphic image is isomorphic to H_α . A decomposition of the central identity into orthogonal rank-one projections gives m_α copies. Such a decomposition is not canonical when $m_\alpha > 1$.

The abstract algebra alone determines the multiplicities, but not the isomorphism classes of the bundles H_α ; these come from its action on G .

13.2. A simple example with a repeated line bundle. Let X have genus at least two and volume one. Choose pairwise nonisomorphic degree-zero line bundles A, B, C , with their Hermitian–Einstein metrics, and put $G = A \oplus B \oplus A \oplus C$, using the same metric on both copies of A . Riemann–Roch gives $h^1(X, L) = g(X) - 1 > 0$ for every nontrivial degree-zero line bundle L . Thus one can choose harmonic forms of unit L^2 norm

$$a \in \Omega^{0,1}(\text{Hom}(B, A)), \quad b \in \Omega^{0,1}(\text{Hom}(A, B)), \quad c \in \Omega^{0,1}(\text{Hom}(C, A)).$$

Set $N_{12} = ta$, $N_{23} = t^2b$, $N_{34} = tc$, and all other entries to zero. These scales come from the scaling exponents $(4, 3, 1, 0)$. An element of $\mathcal{K}$ is

$$x = \begin{pmatrix} x_1 & 0 & u & 0 \\ 0 & x_2 & 0 & 0 \\ v & 0 & x_3 & 0 \\ 0 & 0 & 0 & x_4 \end{pmatrix}, \quad \sum_{i=1}^4 x_i = 0, \quad \|x\|^2 = \sum_i |x_i|^2 + |u|^2 + |v|^2.$$

The six nonzero commutator entries give

$$(77) \quad \begin{aligned} A_t(x) = & t^2(|x_2 - x_1|^2 + |x_4 - x_3|^2 + |u|^2 + |v|^2) \\ & + t^4(|x_3 - x_2|^2 + |v|^2). \end{aligned}$$

This is positive definite on $\mathcal{K}$ for $t > 0$, so Theorem 13.1 proves that E_t is simple. Its upper-triangular filtration has stable degree-zero quotients and a degree-zero line subbundle; hence it is strictly semistable.

The u and v modes have comparison eigenvalues t^2 and $t^2 + t^4$. The diagonal part is the four-vertex path with conductances t^2, t^4, t^2 . Its three positive eigenvalues are

$$2t^2, \quad t^2 + t^4 \pm \sqrt{t^4 + t^8}.$$

Thus there are $2^2 + 1^2 + 1^2 - 1 = 5$ small positive Dolbeault eigenvalues, in increasing order,

$$(78) \quad \begin{aligned} \lambda_1 &= t^4(1 + O(t^2)), \\ \lambda_j &= t^2(1 + O(t^2)) \quad (j = 2, 3), \\ \lambda_j &= 2t^2(1 + O(t^2)) \quad (j = 4, 5). \end{aligned}$$

The full limiting space contains the two off-diagonal maps between the copies of A , in addition to the three trace-free diagonal directions. After adjoining the identity, it is $M_2(\mathbb{C}) \oplus \mathbb{C} \oplus \mathbb{C}$, realized on G .

For completeness the limiting subspaces at each leading coefficient can also be read from this example. The slow line is spanned by $\text{diag}(1, 1, -1, -1)$. The coefficient-one subspace at order t^2 is spanned by E_{13}, E_{31} , and the coefficient-two subspace is spanned by $\text{diag}(1, -1, 0, 0)$ and $\text{diag}(0, 0, 1, -1)$. Indeed, $t^{-2}L_t$ converges to the operator with exactly these eigenspaces and respective eigenvalues 0, 1, 2. In the complementary equation (27), the correction has operator norm $O(t^4)$ by (28), which applies equally to (73). After division by t^2 it tends to zero. Together with (76) and the known multiplicities, this proves convergence of the three spectral projections in every $L^2 \rightarrow C^k$ norm. It does not single out a basis within either two-dimensional limiting subspace.

14. SPECTRAL RECOVERY FOR SEMISTABLE BUNDLES ON CURVES

We now show that the harmonic families of Section 3 can be constructed from every simple strictly semistable bundle on a curve, including those with repeated stable Jordan–Hölder factors. The construction uses the normal form of Lemma 6.1.

14.1. A metric family realizing the graded object. Let E now be simple and strictly semistable. Fix a Jordan–Hölder filtration (41) with stable quotients F_i , allowing repetitions. On a smooth curve a saturated filtration has locally free terms and quotients, so the lemma applies. All quotients have slope $\mu(E)$. Choose Hermitian–Einstein metrics h_i on them, choosing the same metric on isomorphic copies under fixed identifications, and put $h_0 = \bigoplus_i h_i$ on $G = \text{gr}(E) = \bigoplus_i F_i$. Write $G = \bigoplus_\alpha H_\alpha \otimes \mathbb{C}^{m_\alpha}$ as in (70), and set $d = \sum_\alpha m_\alpha^2 - 1$. Apply Lemma 6.1 using these metrics to obtain Φ and β .

Choose any strictly decreasing integers $\ell_1 > \dots > \ell_m = 0$, scaling exponents as in Section 3, and set

$$(79) \quad g_t = \text{diag}(t^{\ell_i} \text{Id}_{F_i}), \quad \Psi_t = g_t \Phi^{-1} : E \longrightarrow G, \quad h_t = \Psi_t^* h_0, \quad t > 0.$$

Define $p_{ij} = \ell_i - \ell_j$ and $p_* = \min_{\beta_{ij} \neq 0} p_{ij}$. This minimum exists: if all entries vanished, E would be a direct sum of $m \geq 2$ stable bundles and hence would not be simple. Let L_t be the operator (73) on $\mathcal{K}_G = H^0(X, \text{End}_0 G)$, with $N_t = \sum_{i < j} t^{p_{ij}} \beta_{ij}$, and let $\nu_j(t)$ be its d eigenvalues in increasing order.

Theorem 14.1 (Recovery along a constructed metric family). *The metrics (79) form an approximate Hermitian–Einstein family on E . More precisely, with $\kappa = 2\pi\mu(E)/V$,*

$$(80) \quad \|K_{h_t} - \kappa \text{Id}_E\|_{L^\infty, \text{op}, h_t} = O(t^{2p_*}).$$

The commutator form L_t is positive definite. Exactly d eigenvalues of the trace-free endomorphism Laplacian tend to zero, all are positive for $t > 0$, and

$$(81) \quad \frac{\lambda_j(h_t)}{\nu_j(t)} = 1 + O(t^{2p_*}), \quad 1 \leq j \leq d, \quad \lambda_{d+1}(h_t) \geq \delta > 0$$

for sufficiently small t .

Let Π_t^E be the orthogonal projection onto these d eigenvalues, and let P_G be the $L^2(h_0)$ projection onto

$$\mathcal{K}_G = H^0(X, \text{End}_0 G) = \left\{ (X_\alpha)_\alpha : \sum_\alpha \text{rk}(H_\alpha) \text{tr } X_\alpha = 0 \right\}.$$

Conjugation by Ψ_t defines a unitary identification $\mathcal{U}_t u = \Psi_t u \Psi_t^{-1}$ of the endomorphism L^2 spaces. For every $k \geq 0$,

$$(82) \quad \|\mathcal{U}_t \Pi_t^E \mathcal{U}_t^{-1} - P_G\|_{L^2(h_0) \rightarrow C^k(h_0)} = O(t^{2p_*}).$$

Adjoining $\mathbb{C}\text{Id}_G$ to the full limiting space gives the algebra $\bigoplus_\alpha M_{m_\alpha}(\mathbb{C})$ acting on G . Its minimal nonzero central idempotents and matrix sizes recover the isotypical components and their multiplicities; the holomorphic images of minimal nonzero idempotents recover the stable factors up to isomorphism.

If the F_i are pairwise nonisomorphic, then $d = m - 1$, L_t is the connected weighted graph Laplacian (20), and the quotient graphs of Theorem 4.1 determine every leading coefficient and corresponding limiting spectral subspace.

Proof. The holomorphic structure transported by Ψ_t is

$$(83) \quad \Psi_t \bar{\partial}_E \Psi_t^{-1} = g_t(\bar{\partial}_0 + \beta)g_t^{-1} = \bar{\partial}_0 + \sum_{i < j} t^{p_{ij}} \beta_{ij}.$$

This is precisely the family of Section 13, with its fixed product metric. The map Ψ_t is a holomorphic isometry from (E, h_t) to that bundle. Naturality of the Chern connection and (32) prove (80); the pointwise operator norms agree under this isometry.

Conjugation by a holomorphic isometry intertwines the Dolbeault operators on endomorphisms and their adjoints. It preserves the trace-free condition and the Hilbert–Schmidt metrics. Thus $\mathcal{U}_t$ identifies the endomorphism Laplacians unitarily. Since E is simple, Theorem 13.1 makes L_t positive definite and gives (81), with $\delta = \gamma/4$ in the notation of that theorem. The same theorem proves (82). When all multiplicities equal one, Theorems 3.1 and 4.1 give connectedness and the graph contraction description of individual decay orders and limiting spectral subspaces.

Finally, stability of the distinct representatives gives

$$\mathcal{K}_G \oplus \mathbb{C}\text{Id}_G = H^0(X, \text{End } G) = \bigoplus_\alpha \text{Id}_{H_\alpha} \otimes M_{m_\alpha}(\mathbb{C}).$$

The central idempotent for α has image $H_\alpha \otimes \mathbb{C}^{m_\alpha}$. Its matrix ideal has dimension m_α^2 , and a minimal nonzero idempotent in that ideal has image isomorphic to H_α . Decomposing the central identity into orthogonal rank-one projections recovers m_α copies. Their direct sum

over α is $\text{gr}(E)$. The Jordan–Hölder theorem makes the isomorphism class of this graded bundle independent of the chosen filtration [Kob87]. $\square$

The construction depends on a filtration, quotient metrics, and integer scaling exponents; the scope of such statements is summarized in Section 1.6.

15. A SLOPE INEQUALITY AND ITS SHARPNESS

This section proves Theorem 1.3 and shows that its constants are optimal in every rank.

15.1. Projections associated with coherent subsheaves. We recall the analytic input that permits the use of a possibly singular subsheaf as a test section. The Sobolev spaces below are defined using any smooth background connection on $\text{End } E$.

Lemma 15.1 (Subsheaf degree formula). *Let $\mathcal{S} \subset E$ be a saturated coherent subsheaf of rank s . Outside an analytic subset $Z \subset X$ of complex codimension at least two, the sheaf $\mathcal{S}$ is a holomorphic subbundle of E . Its h -orthogonal projection $P_{\mathcal{S}}$ extends as an element of $W^{1,2}(X, \text{End } E)$ and satisfies, almost everywhere,*

$$P_{\mathcal{S}}^2 = P_{\mathcal{S}} = P_{\mathcal{S}}^{*h}, \quad \text{tr } P_{\mathcal{S}} = s, \quad (\text{Id}_E - P_{\mathcal{S}})\bar{\partial}_{\text{End } E} P_{\mathcal{S}} = 0.$$

Moreover,

$$(84) \quad 2\pi \deg_{\omega} \mathcal{S} = \int_X \text{tr}(K_h P_{\mathcal{S}}) dV_{\omega} - \|\bar{\partial}_{\text{End } E} P_{\mathcal{S}}\|_{L^2}^2.$$

Proof. The sheaves $\mathcal{S}$ and $E/\mathcal{S}$ are torsion-free, so both are locally free away from an analytic subset of codimension at least two. On this complement, a local splitting of the resulting exact sequence makes $\mathcal{S}$ a subbundle. The projection identities follow there from orthogonality and holomorphicity.

For a smooth subbundle, the Gauss–Codazzi formula gives (84). With the Hilbert–Schmidt convention used here, $\|\bar{\partial}_{\text{End } E} P_{\mathcal{S}}\|_{L^2}^2$ is the squared L^2 norm of the second fundamental form. The factor 2π follows from $c_1 = \frac{\sqrt{-1}}{2\pi} \text{tr } F$ and the contraction identity

$$\alpha \wedge \frac{\omega^{n-1}}{(n-1)!} = (\Lambda_{\omega} \alpha) \frac{\omega^n}{n!}$$

for a $(1,1)$ -form α .

For a saturated coherent subsheaf, we use the standard extension of this formula in the theory of weakly holomorphic projections [UY86, Kob87]. In particular, the projection is bounded and its second fundamental form is square-integrable. A regularization argument establishing this integrability and preservation of the degree is given in [Jac14, Sections 2–3]. Saturation ensures that there is no divisorial contribution from the singular set. The regularized degree formula consequently gives (84) on $X \setminus Z$.

For completeness, square-integrability of the second fundamental form also gives the required Sobolev statement. Since $P_{\mathcal{S}}$ is self-adjoint, its $(1,0)$ covariant derivative is the adjoint of its $(0,1)$ derivative, so its full first covariant derivative is square-integrable on $X \setminus Z$. An analytic set of complex codimension at least two has zero $W^{1,2}$ capacity. Applying cutoff functions with Dirichlet energy tending to zero, and using boundedness of $P_{\mathcal{S}}$, extends it across Z as a $W^{1,2}$ section. The integral formula is unchanged because Z has measure zero. $\square$

Remark 15.2. The condition $(\text{Id}_E - P_{\mathcal{S}})\bar{\partial}_{\text{End } E} P_{\mathcal{S}} = 0$ says that the image of $P_{\mathcal{S}}$ is holomorphic. It does not say that $P_{\mathcal{S}}$ is a holomorphic endomorphism. The latter would require $\bar{\partial}_{\text{End } E} P_{\mathcal{S}} = 0$ and would give a holomorphic direct-sum decomposition of E .

The following elementary estimate uses the trace-free condition on the curvature, in addition to the spectrum of the projection.

Lemma 15.3. *Let A be a self-adjoint trace-free endomorphism of an r -dimensional Hermitian vector space, and let Π be an orthogonal projection of rank s , where $0 < s < r$. Set $q = \Pi - \frac{s}{r} \text{Id}$. Then*

$$(85) \quad |q|^2 = \frac{s(r-s)}{r}, \quad |\text{tr}(Aq)| \leq \min\{s, r-s\} \|A\|_{\text{op}}.$$

The coefficient $\min\{s, r-s\}$ in the second inequality is optimal among such matrices and projections.

Proof. The eigenvalues of q are $(r-s)/r$, with multiplicity s , and $-s/r$, with multiplicity $r-s$, which proves the norm identity. Since $\text{tr } A = 0$,

$$\text{tr}(Aq) = \text{tr}(A\Pi) = -\text{tr}(A(\text{Id} - \Pi)).$$

Estimating the two compressed traces by their respective dimensions gives (85).

To see optimality, suppose first that $s \leq r-s$ and take $A = \text{Id}$ on $\text{im } \Pi$ and $A = -\frac{s}{r-s} \text{Id}$ on its orthogonal complement. Then $\text{tr } A = 0$, $\|A\|_{\text{op}} = 1$, and $\text{tr}(Aq) = s$. For $s \geq r-s$, take $A = \frac{r-s}{s} \text{Id}$ on $\text{im } \Pi$ and $A = -\text{Id}$ on its complement. $\square$

15.2. Proof of the slope estimate.

Proof of Theorem 1.3. Let P_S be the projection supplied by Lemma 15.1, and put

$$q = P_S - \frac{s}{r} \text{Id}_E.$$

This is a trace-free $W^{1,2}$ endomorphism and $\bar{\partial}_{\text{End } E} q = \bar{\partial}_{\text{End } E} P_S$. Subtracting (84) from s/r times the degree formula for E gives

$$(86) \quad 2\pi s(\mu_\omega(E) - \mu_\omega(\mathcal{S})) = \|\bar{\partial}_{\text{End } E} q\|_{L^2}^2 - \int_X \text{tr}(K_h^0 q) dV_\omega.$$

The scalar part of K_h disappears because $\text{tr } q = 0$ pointwise; no assumption on $\text{tr } K_h$ is used.

The Rayleigh inequality in (1) extends to trace-free $W^{1,2}$ sections by density. Explicitly, a smooth approximation may be made trace-free by subtracting its scalar part. Hence Lemma 15.3 gives

$$(87) \quad \|\bar{\partial}_{\text{End } E} q\|_{L^2}^2 \geq \lambda_1(h) \|q\|_{L^2}^2 = \lambda_1(h) V \frac{s(r-s)}{r}.$$

The same lemma, applied pointwise to $A = K_h^0$, gives

$$(88) \quad \left| \int_X \text{tr}(K_h^0 q) dV_\omega \right| \leq V \min\{s, r-s\} \varepsilon_1(h).$$

Combining (86)–(88) and dividing by $2\pi s$ proves (5), since

$$\frac{r \min\{s, r-s\}}{s(r-s)} = \frac{r}{\max\{s, r-s\}}.$$

Finally,

$$\max_{1 \leq s \leq r-1} \frac{r}{\max\{s, r-s\}} = \frac{r}{\lceil r/2 \rceil} = c_r.$$

Under (6), every saturated proper subsheaf has strictly smaller slope. An arbitrary subsheaf has slope at most that of its saturation, so E is stable. $\square$

Remark 15.4. The coefficient for all subsheaf ranks is $c_r = 2$ in even rank and $c_r = 2r/(r+1)$ in odd rank. For example, rank three gives $c_3 = 3/2$. For rank four the coefficient in (5) is $4/3$ for subsheaves of rank one or three, and 2 for subsheaves of rank two. Pointwise optimality in Lemma 15.3 does not by itself give optimality of the geometric estimate, which requires simultaneous equality in the curvature estimate and the Rayleigh inequality. Corollary 15.9 shows that both become asymptotically sharp on suitable nonsplit extensions. Every coefficient in (5), and c_r itself, is optimal in every rank.

15.3. Consequences.

Corollary 15.5 (A uniform slope margin). *If $\gamma(h) = \lambda_1(h) - c_r \varepsilon_1(h) > 0$, then every coherent subsheaf $\mathcal{S} \subset E$ of rank $0 < s < r$ satisfies*

$$\mu_\omega(E) - \mu_\omega(\mathcal{S}) \geq \frac{V(r-s)}{2\pi r} \gamma(h) \geq \frac{V}{2\pi r} \gamma(h).$$

In particular, if E is simple and $K_h^0 = 0$, then

$$(89) \quad \mu_\omega(E) - \mu_\omega(\mathcal{S}) \geq \frac{V(r-s)}{2\pi r} \lambda_1(h) > 0.$$

Proof. For a saturated subsheaf, replace the rank-dependent coefficient in (5) by c_r . For any other subsheaf, apply the result to its saturation, which has the same rank and at least the same degree. If $K_h^0 = 0$, use (5) directly and the positivity of $\lambda_1(h)$ for a simple bundle. $\square$

Remark 15.6. An irreducible Hermitian–Einstein connection has only scalar holomorphic endomorphisms by the Bochner formula, so (89) applies in the classical setting. Conversely, a stable bundle admits a Hermitian–Einstein metric by [Don85, UY86], and that metric satisfies (6). Thus existence of a metric satisfying the criterion is equivalent to stability. The existence direction here is the classical correspondence; the estimate does not construct such a metric from an arbitrary one.

Corollary 15.7 (Spectral collapse). *Suppose that E is simple and strictly semistable. Fix a saturated subsheaf $\mathcal{S} \subset E$ of rank $0 < s < r$ with $\mu_\omega(\mathcal{S}) = \mu_\omega(E)$. Then every smooth Hermitian metric h satisfies*

$$(90) \quad 0 < \lambda_1(h) \leq \frac{r}{\max\{s, r-s\}} \varepsilon_1(h) \leq c_r \varepsilon_1(h).$$

Consequently, if h_j is any sequence with $\varepsilon_1(h_j) \rightarrow 0$, then $\lambda_1(h_j) \rightarrow 0$.

Proof. Strict semistability provides an equal-slope proper subsheaf. Its saturation still has equal slope, by semistability. Substituting this subsheaf in (5) proves the upper bound. Simplicity supplies strict positivity for each fixed metric. $\square$

Semistable bundles on compact Kähler manifolds admit approximate Hermitian–Einstein metrics [Jac14]. In the normalizations of this paper, these are metrics satisfying

$$\|K_{h_j} - \kappa \operatorname{Id}_E\|_{L^\infty, \text{op}} \rightarrow 0, \quad \kappa = \frac{2\pi\mu_\omega(E)}{V}.$$

Their trace-free mean curvatures tend to zero, so Corollary 15.7 applies to every such sequence on a simple strictly semistable bundle. The conclusion also holds under the weaker assumption in the corollary, without convergence of the scalar part of the mean curvature. For a nonsimple bundle, the lowest eigenvalue in (1) is already zero for every metric; no conclusion about its first positive eigenvalue is asserted.

Proposition 15.8 (Scalar changes of metric). *If f is a smooth real function on X and $\tilde{h} = e^f h$, then*

$$\lambda_1(\tilde{h}) = \lambda_1(h), \quad K_{\tilde{h}}^0 = K_h^0, \quad \varepsilon_1(\tilde{h}) = \varepsilon_1(h).$$

Proof. Multiplication of h by a scalar function leaves the induced metric on $\text{End } E$ unchanged, and the Dolbeault operator is fixed. It therefore leaves both norms in (1) unchanged. The Chern curvatures of $\tilde{h}$ and h differ by a scalar $(1, 1)$ -form times Id_E , so their trace-free mean curvatures agree. The operator norm of an endomorphism is also unchanged by a scalar change of metric. $\square$

15.4. Sharpness in every rank.

Corollary 15.9 (Geometric sharpness in every rank). *Let X be a curve of genus at least two, and let $r \geq 2$ and $1 \leq s \leq r - 1$. There are a simple strictly semistable bundle E of rank r with a saturated subbundle of rank s and slope $\mu(E)$, and metrics h_t on E with $\varepsilon_1(h_t) \rightarrow 0$, such that*

$$\frac{\lambda_1(h_t)}{\varepsilon_1(h_t)} \rightarrow \frac{r}{\max\{s, r - s\}}, \quad \frac{\lambda_1(h_t)}{\|K_{h_t}^0\|_{L^\infty, \text{op}}} \rightarrow \frac{r}{\max\{s, r - s\}}.$$

Consequently the coefficient $r/\max\{s, r - s\}$ in (5) cannot be decreased for any (r, s) , and the constant c_r in (6) is optimal in every rank, also for the criterion with the L^∞ norm in place of ε_1 .

Proof. Choose stable F of rank $a = s$ and G of rank $b = r - s$, both of degree zero, and nonisomorphic. A nonzero map between them would be an isomorphism, so $h^0(\text{Hom}(G, F)) = 0$. By Riemann–Roch, $h^1(\text{Hom}(G, F)) = ab(g - 1) > 0$, so there is a nonzero harmonic β .

With $v = Rz$, consider on E_t the metric $h'_t = h_0 e^{-t^2 v}$, where $h'_t(x, y) = h_0(e^{-t^2 v} x, y)$. As in the proof of Lemma 7.2, the first variation of mean curvature at h_0 is Δ_0 . For $t \neq 0$ the operator changes by $O(t)$. Hence

$$K_{h'_t} = \kappa \text{Id} + t^2(C - \Delta_0 v) + O_{C^0}(t^3) = \kappa \text{Id} + ct^2 q + O_{C^0}(t^3),$$

since $\Delta_0 v = z = C - cq$. The endomorphism $K_{h'_t}^0$ is h'_t -self-adjoint, so its operator norm is its spectral radius. The eigenvalues of $ct^2 q$ are $ct^2 b/r$ and $-ct^2 a/r$. Therefore

$$\|K_{h'_t}^0\|_{\text{op}} = \frac{ct^2 \max\{a, b\}}{r} + O(t^3)$$

uniformly on X , and the same holds for ε_1 and the L^∞ norm. Since $e^{-Ct^2} h_0 \leq h'_t \leq e^{Ct^2} h_0$, (48) and Theorem 2.1 give $\lambda_1(h'_t) = ct^2(1 + O(t^2))$. The ratios therefore tend to $r/\max\{a, b\}$.

Transport by g_t in (16). This gives the metrics $g_t^* h'_t$ on the fixed bundle $E = E_1$, with the same λ_1 , ε_1 , and L^∞ norm. The subbundle $F \subset E$ has rank s and equal slope, so (5) gives $\lambda_1 \leq (r/\max\{s, r - s\})\varepsilon_1$ for every metric on E . A smaller coefficient would contradict the limit. For $s = \lfloor r/2 \rfloor$ the coefficient is c_r . $\square$

Remark 15.10. The metrics h'_t make both estimates in the proof of Theorem 1.3 asymptotically sharp. The Rayleigh bound is sharp because $u_t \rightarrow q/\sqrt{V_q}$. The trace-free matrix estimate is sharp because the correction replaces C by its projection cq , which is constant on each block with the larger absolute value on the smaller block.

Without the correction, on a curve, $\lambda_1/\varepsilon_1 \rightarrow (r/ab) \int \sum_k \sigma_k^2 / \int \sigma_{\max}^2$, where the σ_k are the pointwise singular values of β . This reaches $r/\max\{a, b\}$ exactly when the singular values agree almost everywhere. It happens, for example, for line-bundle extensions, when $\min\{a, b\} = 1$, and for $\beta = \gamma \otimes \text{Id}_W$ on $F = L_1 \otimes W$, $G = L_2 \otimes W$, with $L_1 L_2^{-1}$ chosen so that

$W \otimes L_1 L_2^{-1} \not\cong W$. It does not give the L^∞ statement: in rank two, $|\beta|^2$ vanishes somewhere, since $\bar{\beta}$ corresponds to a holomorphic section of a line bundle of positive degree.

For line-bundle extensions, put $f = |\beta|^2 - V^{-1} \|\beta\|_{L_2}^2$ and let R_{sc} be the inverse of the scalar $\bar{\partial}^* \bar{\partial}$ on mean-zero functions. Since $z = 2fq$ and q is parallel, $d = \frac{4}{V} \langle f, R_{\text{sc}} f \rangle$. More generally, $d = 0$ exactly when $C = cq$. In that case q is an exact eigensection with eigenvalue ct^2 , which is the first eigenvalue for small $|t|$.

16. OPEN PROBLEMS

- (1) *Arbitrary approximate Hermitian–Einstein sequences.* Corollary 15.7 concerns eigenvalues only. For a general sequence the Hilbert norms and adjoints depend on h_j , and a statement about eigensections must specify a gauge, the limiting holomorphic structure, and the topology of convergence.
- (2) *The degenerate case.* When a tight edge carries a vanishing multiplier, iterated logarithms can appear. Theorem 9.5 reduces the question to the diagonal flow, but the sharp asymptotics are known here only for the zigzag of Theorem 11.4.
- (3) *Individual coefficients.* At each faster rate we show that the sum of the coefficients depends on the initial metric. Each coefficient does so when the edges of that exponent form a forest on the tight components, parallel edges counting as cycles: prescribing the constants Z_C along the forest scales all these conductances by a common factor. The general case is open.
- (4) *Repeated factors along the flow.* Then the diagonal flow is replaced by a moment-map flow for $\prod_\alpha GL(m_\alpha)$. Lam [Lam26] gives the metric asymptotics up to a bounded term; the spectral questions are open.
- (5) *More than two steps in dimension at least two.* The Maurer–Cartan equation requires cup-product corrections, and the matching argument needs a Kuranishi-slice version.
- (6) *Singular graded objects,* where the flow bubbles [SW15].
- (7) *Variation of the polarization.* Clarke–Tipler [CT23] study convergence of Hermitian Yang–Mills connections to the graded bundle as the polarization varies. Our fixed-polarization calculation gives no effective wall-crossing radius.

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BERND JOHANNES WUEBBEN, NEW YORK, NY, wuebben@gmail.com